

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf\_3d035317-127\agent-aaedd96d5dad9c875.jsonl`
- SHA-256: `d5d38fb83996ba1bcab7889eefcec9752aa752fba73945ab022a7148979e8bfc`
- Source modified: `2026-06-22T09:51:11+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf\_3d035317-127`
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`

Transcript

1. user (2026-06-22T09:46:23.311Z)

You are fixing staleness in ONE doc of the KhelSutra repo, applying an audit that already ran. Be SURGICAL: minimal, targeted edits that match the audited fix; preserve the doc's voice, structure, and markdown formatting. Do NOT wholesale-rewrite a doc.

KhelSutra repo root: C:\Users\avidu\Projects\kheelsutra
ENGINE repo root (sibling, for cross-repo verification): C:\Users\avidu\Projects\badminton-highlight-indexer (origin/master freshly fetched)
Today: 2026-06-22. KhelSutra is on branch docs/freshness-sweep (off master).

YOUR DOC (edit ONLY this file): C:\Users\avidu\Projects\kheelsutra\docs\PER_RALLY_SELECTION.md

The audit findings are in this JSON file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\tasks\w3dbzrpbby.output
Shape: { result: { perDoc: [{doc, overallStatus, findings, summary}], staleFindings: [{doc, claim, docLocation, verdict, evidence, suggestedFix, severity}] } }.
Read it and select ONLY entries where doc === "docs/PER_RALLY_SELECTION.md". Your worklist = that doc's staleFindings. Also skim that doc's perDoc.findings for "unverifiable" items ? do NOT act on those unless you can now positively verify them.

For each stale finding:

1. RE-VERIFY it yourself against the LIVE repo before editing ? Read the doc lines + the cited code. For any claim about the ENGINE (paths like backend/main.py, /api/*, native_wasb_runner.py, auth/CORS), verify against the engine repo's origin/master: e.g. `git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "<symbol>" origin/master -- backend/` or `git -C ... show origin/master:backend/main.py`. Trust nothing blindly ? the audit itself may be slightly off and engine line numbers DRIFT between PRs.
2. If confirmed, apply the suggestedFix as a SURGICAL Edit. For brittle ENGINE line-number anchors, prefer SYMBOL-based references (e.g. `backend/main.py::get\_config`) over chasing exact line numbers ? but only after confirming the symbol exists on engine origin/master.
3. DOCS ONLY: if a fix would require a CODE change (e.g. a comment in site/server.js), do NOT edit code ? record it under codeChangesNeeded.

Hard rules:

- Edit ONLY your assigned doc file. Run NO git WRITE commands (no add/commit/checkout/branch/stash) ? read-only git for verification only.
- Keep status markers HONEST: flip Todo->Done only WITH the PR number / file evidence; never invent a completion.
- If a finding is wrong, not reproducible, or too risky to auto-apply, SKIP it and explain under skipped.
- Do not "improve" beyond the findings; no scope creep.

Return the structured report.

2. assistant (2026-06-22T09:46:26.174Z)

I'll start by reading the audit findings and the target doc.

3. user (2026-06-22T09:46:28.551Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-591 of 1722 total (61859 tokens, cap 25000). Call Read with offset=592 limit=591 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      {
2      "summary": "Audit every KhelSutra doc for staleness against the live repo, returning
structured findings",
3      "agentCount": 15,
4      "logs": [
5      "audited 15/15 docs; 44 stale findings; archive-candidates: none"
6      ],
7      "result": {
8      "perDoc": [
9      {
10     "doc": "NEXT_STEPS.md",
11     "overallStatus": "minor-staleness",
12     "summary": "NEXT_STEPS.md is overwhelmingly accurate against master: routes,
file paths, the Worker/D1 names, the 6-language localization, the 3-step wizard, the per-rally
PR numbers (#11/#14/#21/#28/#30/#31), the appliance/per-rally doc statuses, and every still-
pending \"Queued\" item all check out against code and git. The only real staleness is the \"PRs
through #93 merged\" count, which lags by one because PR #94 (the very commit that wrote
\"through #93\") merged afterward; plus a low-severity GCS-artifact filename mismatch. Not an
archive candidate ? it is the live next-steps tracker.",
13     "findings": [
14     {
15     "claim": "PRs through #93 merged (master).",
16     "docLocation": "NEXT_STEPS.md:13",
17     "verdict": "stale",
18     "evidence": "git log --oneline -1 = '50b2bd7 chore(repo): ... correct stale
NEXT_STEPS claims (#94)'; total 94 commits/PRs on master. Ironically #94 is the commit that
updated this line to '#93', then merged itself, so the doc now lags one PR behind master.",
19     "suggestedFix": "Update to 'PRs through #94 merged (master).'",
20     "severity": "medium"
21     },
22     {
23     "claim": "Where we are (2026-06-20): ... khelsutra.guru is LIVE and fully
shipped (landing + /capture checklist + OG image + segmented Request-Access form).",
24     "docLocation": "NEXT_STEPS.md:5-14",
25     "verdict": "correct",
26     "evidence": "site/server.js:98 maps /capture->capture.html;
site/public/og.png + capture.html tracked; index.html:191-193 textareas + lines 537-581
academy/individual/gpu/storage fields. Header date 2026-06-20 is slightly behind the latest work
(PR #94 merged 2026-06-22) but the substance holds.",
27     "suggestedFix": "Optionally bump the 'Where we are' date to 2026-06-22 to
match the latest merged work.",
28     "severity": "low"
29     },
30     {
31     "claim": "On Cloudflare (Worker hidden-king-khelsutra + D1 khelsutra-
access): the form persists to D1 and emails a summary via an Email Routing send binding.",
32     "docLocation": "NEXT_STEPS.md:7-9",
33     "verdict": "correct",
34     "evidence": "site/wrangler.toml: name = \"hidden-king-khelsutra\";
[[d1_databases]] database_name = \"khelsutra-access\"; [[send_email]] name = \"NOTIFY\".
site/src/index.js:80 handles POST /api/request-access -> insert into D1 binding DB.",
35     "suggestedFix": "",
36     "severity": "low"
37     },
38     ]
39     },
40     }
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38         {
39             "claim": "CI gates every PR ? three jobs: full site npm test, web/ SPA build
+ coverage floor, and a real-browser layout check.",
40             "docLocation": "NEXT_STEPS.md:9-12",
41             "verdict": "correct",
42             "evidence": "site/package.json scripts include test (node --test),
test:layout (layout.browser.mjs); web has a coverage floor (#80). PR #94 message confirms 'three
jobs: site npm test = 86 green, web SPA build+coverage, real-browser layout'. Non-browser site
suite ran green (61 tests in the 4 non-layout files).",
43             "suggestedFix": "",
44             "severity": "low"
45         },
46         {
47             "claim": "The form also runs on a dependency-free Node + SQLite host
(site/server.js), live on the DO droplet -> http://168.144.126.176:8787 (systemd khelsutra-
site).",
48             "docLocation": "NEXT_STEPS.md:11-13",
49             "verdict": "correct",
50             "evidence": "site/server.js exists, uses node:sqlite (header comment line
2). IP 168.144.126.176 + port 8787 + 'khelsutra-site' all appear in
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48 ('/srv/khelsutra-site, node :8787') and deploy refs.
Live-status is external/unverifiable from the repo, but the host code and identifiers are
correct.",
51             "suggestedFix": "",
52             "severity": "low"
53         },
54         {
55             "claim": "Docs: docs/REQUEST_ACCESS_FORM.md, docs/PORTABILITY.md.",
56             "docLocation": "NEXT_STEPS.md:13-14",
57             "verdict": "correct",
58             "evidence": "git ls-files: both docs/REQUEST_ACCESS_FORM.md and
docs/PORTABILITY.md are tracked.",
59             "suggestedFix": "",
60             "severity": "low"
61         },
62         {
63             "claim": "DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22): localized to
all 6 languages (en/kn/hi/es/da/id); full record docs/archives/FRIENDLY_AND_MULTILINGUAL.md.",
64             "docLocation": "NEXT_STEPS.md:17-21",
65             "verdict": "correct",
66             "evidence": "docs/archives/FRIENDLY_AND_MULTILINGUAL.md is tracked; git log
shows PR #93 'docs: archive the Friendly & Multilingual tracker'. web/src/i18n/index.ts:25
SUPPORTED_LOCALES = [en,kn,hi,es,da,id]; PR #87 added es/da/id to the marketing site.",
67             "suggestedFix": "",
68             "severity": "low"
69         },
70         {
71             "claim": "Interactive per-rally selection MVP/M0 COMPLETE; Done: P0 design
(#11), P1 scaffold (#14), P2 grid+hook (#21), P3 parseQuery query bar (#28), P4 SelectionFooter
(#30), P5 real load+download (#31). Tracked doc docs/PER_RALLY_SELECTION.md (IN PROGRESS).",
72             "docLocation": "NEXT_STEPS.md:32-41",
73             "verdict": "correct",
74             "evidence": "git log: cb74ce5 #11 'tracked design ... RallyPicker'; aac7e0c
#14 'P1 scaffold'; 62598e7 #21 'P2 selectable rally grid + hook'; d537d19 #28 'P3 deterministic
parseQuery() query bar'; aff8de8 #30 'P4 SelectionFooter + min_shot_count warning'; 8bf200d #31
'P5 load real rallies + Build->compile->preview->download'. docs/PER_RALLY_SELECTION.md status
header = 'IN PROGRESS ? MVP/M0 (P1-P5) COMPLETE'.",
75             "suggestedFix": "",
76             "severity": "low"
77         },
78         {
79             "claim": "P2c first-run setup wizard ? 3-step SetupWizard from
/api/capabilities, 6 languages.",
80             "docLocation": "NEXT_STEPS.md (implied via track) and verified for the
merged #88",

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81         "verdict": "correct",
82         "evidence": "web/src/components/SetupWizard.tsx:22 'const STEPS = 3'; line 1
comment 'Three quick steps'; line 3 'Localized (en/kn/hi/es/da/id)'; web/src/api/client.ts:139
GET /capabilities. git: 7ae9cee #88.",
83         "suggestedFix": "",
84         "severity": "low"
85     },
86     {
87         "claim": "Single-box appliance ? Tracked design
docs/LOCAL_APPLIANCE_ARCHITECTURE.md (DRAFT, owner-gated).",
88         "docLocation": "NEXT_STEPS.md:52-58",
89         "verdict": "correct",
90         "evidence": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md status header: 'Status:
DRAFT ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5 bridge)'.",
91         "suggestedFix": "",
92         "severity": "low"
93     },
94     {
95         "claim": "Queued: Always-Use-HTTPS + HSTS (Cloudflare) and Cloudflare Web
Analytics token paste into site/public/index.html ? both still pending.",
96         "docLocation": "NEXT_STEPS.md:44-45",
97         "verdict": "correct",
98         "evidence": "No Strict-Transport-Security/HSTS header found in site/*.js or
wrangler.toml. site/public/index.html:629 still has data-cf-beacon token 'REPLACE_WITH_CF_TOKEN'
(commented-out block). Both correctly listed as not-yet-done.",
99         "suggestedFix": "",
100        "severity": "low"
101    },
102    {
103        "claim": "Droplet -> production: add VM-path SMTP so the Node host also
emails (currently noopNotify).",
104        "docLocation": "NEXT_STEPS.md:46-47",
105        "verdict": "correct",
106        "evidence": "site/server.js:14 imports noopNotify; line 24 'const notify =
noopNotify(); // VM host: no email yet (add SMTP later)'.",
107        "suggestedFix": "",
108        "severity": "low"
109    },
110    {
111        "claim": "Org consolidation DONE; khelsutra (this site repo) transfer is an
owner step still PENDING (unchecked).",
112        "docLocation": "NEXT_STEPS.md:65-73",
113        "verdict": "correct",
114        "evidence": "git remote -v shows origin =
https://github.com/avidullu/khelsutra.git (NOT under Khelsutra org) ? consistent with the '?
owner step pending' marker. Other repos already moved: docs/DRY_RUN_CUJ.md:49 links
github.com/Khelsutra/badminton-highlight-indexer#289.",
115        "suggestedFix": "",
116        "severity": "low"
117    },
118    {
119        "claim": "A real-UI CUJ screencast exists -> promo_cuj.mp4/.gif in
gs://khelsutra-rally-corpus/promo/.",
120        "docLocation": "NEXT_STEPS.md:29-31",
121        "verdict": "unverifiable",
122        "evidence": "No GCS access to confirm the object. The companion doc
docs/DRY_RUN_CUJ.md:85 names the actual artifacts 'khelsutra-bucket-cuj-screencast.mp4' and ':60
khelsutra-i18n-screencast.mp4' in the same gs://khelsutra-rally-corpus/promo/ prefix;
'promo_cuj.mp4' appears nowhere else in the repo, suggesting a possible filename mismatch.",
123        "suggestedFix": "Reconcile the filename with docs/DRY_RUN_CUJ.md (likely
'khelsutra-bucket-cuj-screencast.mp4') or confirm 'promo_cuj.mp4' is a real object in the
bucket.",
124        "severity": "low"
125    },
126    {

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127         "claim": "Control plane (control-plane/) at beta.",
128         "docLocation": "NEXT_STEPS.md:62",
129         "verdict": "correct",
130         "evidence": "control-plane/ directory does not exist in the repo (test -d
looking/aspirational item, not a claim of present existence ? so it is not stale as written.",
131         "suggestedFix": "",
132         "severity": "low"
133     }
134 ]
135 },
136 {
137     "doc": "README.md",
138     "overallStatus": "significant-staleness",
139     "summary": "The repo-structure list, deploy references, and canonical-doc
pointers are all correct, but the README's headline status is badly out of date: it still calls
the repo a \"pre-alpha scaffold\" whose SPA has yet to land \"per milestone M1\", whereas the
web/ SPA is in fact built, multilingual, and shipped (per-rally MVP P0?P5, P2c wizard, in-UI
upload, 6 languages; khelsutra.guru is LIVE). The control-plane/ directory it lists also does
not yet exist.",
140     "findings": [
141     {
142         "claim": "Status: pre-alpha scaffold. SPA lands in `web/` per execution-plan
milestone M1.",
143         "docLocation": "README.md:9",
144         "verdict": "stale",
145         "evidence": "web/src/ has 51 tracked source files incl. web/src/App.tsx;
`git log --oneline -- web/` = 31 commits; merged PRs show the SPA is feature-complete: P2c
first-run wizard (#88), in-UI file upload T4 (#66), per-rally selection MVP P0?P5
(#11/#14/#21/#28/#30/#31), 6-language catalogs (#67). NEXT_STEPS.md:7 states 'khelsutra.guru is
LIVE and fully shipped'. UI_ALPHA_EXECUTION_PLAN.md:36 defines M1 only as the initial
'Vite+React+TS+Tailwind scaffold', which is long past.",
146         "suggestedFix": "Replace with a current status, e.g.: 'Status: alpha ? the
web/ SPA is built and multilingual (6 languages), with the per-rally selection MVP and first-run
setup wizard shipped; khelsutra.guru / app.khelsutra.guru are live (Cloudflare Pages +
cloudflared tunnel + Access). See NEXT_STEPS.md.'",
147         "severity": "high"
148     },
149     {
150         "claim": "the web app (alpha SPA) now, the hosted control plane at beta",
151         "docLocation": "README.md:4",
152         "verdict": "stale",
153         "evidence": "This tagline pairs with line 9's 'pre-alpha scaffold', which is
contradicted by the shipped, live SPA (see web/ evidence above and NEXT_STEPS.md:7). The 'alpha
SPA now' framing is directionally right but undercut by the stale 'pre-alpha scaffold' status it
sits beside.",
154         "suggestedFix": "Keep 'web app (alpha SPA) now' but align the rest of the
intro/status so it no longer describes the SPA as not-yet-landed.",
155         "severity": "medium"
156     },
157     {
158         "claim": "`control-plane/` ? (beta) Cloud Run FastAPI control plane",
159         "docLocation": "README.md:17",
160         "verdict": "stale",
161         "evidence": "No control-plane/ directory exists: `ls control-plane` -> 'No
such file or directory'; `git ls-files control-plane/` returns nothing. Top-level tracked dirs
are deploy/, docs/, locales/, site/, web/ only.",
162         "suggestedFix": "Mark it as a future/planned directory that does not yet
exist (e.g. move it out of the 'Structure' list into a 'Planned (beta)' note), so a reader
doesn't expect the path to be present.",
163         "severity": "medium"
164     },
165     {
166         "claim": "Engine ... sibling repo `badminton-highlight-indexer` (rename
pending).",

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167         "docLocation": "README.md:6",
168         "verdict": "correct",
169         "evidence": "Sibling repo still exists under that name: `ls
C:/Users/avidu/Projects/` shows 'badminton-highlight-indexer'; no rename has occurred, so
'rename pending' holds.",
170         "suggestedFix": "",
171         "severity": "low"
172     },
173     {
174         "claim": "Canonical planning docs live in the engine repo's docs/ for now:
UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md.",
175         "docLocation": "README.md:7-8",
176         "verdict": "correct",
177         "evidence": "All three exist in the engine repo (badminton-highlight-
indexer/docs/UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md present) and are NOT
tracked in khelsutra/docs (`git ls-files docs/` in khelsutra returns none of them; `ls
docs/UI_PROPOSAL.md` -> not found).",
178         "suggestedFix": "",
179         "severity": "low"
180     },
181     {
182         "claim": "`deploy/` ... alpha deploy kit ... DEPLOY_ALPHA.md ... +
cloudflared.config.example.yml",
183         "docLocation": "README.md:15",
184         "verdict": "correct",
185         "evidence": "git ls-files deploy/ shows deploy/DEPLOY_ALPHA.md and
deploy/cloudflared.config.example.yml (plus deploy/droplet/*). DEPLOY_ALPHA.md:1-5 matches the
described 'shareable page, no SSH; SPA on Pages + engine on api.* tunnel + CF Access' content.",
186         "suggestedFix": "",
187         "severity": "low"
188     },
189     {
190         "claim": "`site/` ? public marketing landing page (static; served at
khelsutra.guru)",
191         "docLocation": "README.md:13",
192         "verdict": "correct",
193         "evidence": "site/ exists with site/public/index.html, server.js, etc.;
DEPLOY_ALPHA.md and NEXT_STEPS.md confirm the marketing site is served at khelsutra.guru (apex)
while the SPA is at app.khelsutra.guru.",
194         "suggestedFix": "",
195         "severity": "low"
196     },
197     {
198         "claim": "`web/` ? React SPA (alpha; Vite + React + TS + Tailwind;
Cloudflare Pages build root)",
199         "docLocation": "README.md:14",
200         "verdict": "correct",
201         "evidence": "web/ is a Vite+React+TS+Tailwind SPA (web/README.md:3 'Vite +
React + TS + Tailwind v4'); DEPLOY_ALPHA.md:14 confirms 'app.khelsutra.guru -> Cloudflare Pages
(the web/ SPA)'. The tech-stack/build-root facts are accurate; only the inline 'alpha' maturity
ties into the stale status above.",
202         "suggestedFix": "",
203         "severity": "low"
204     },
205     {
206         "claim": "License: All rights reserved (private; no OSS license).",
207         "docLocation": "README.md:19-21",
208         "verdict": "correct",
209         "evidence": "No LICENSE file is tracked (not present in `git ls-files`);
consistent with a private 'all rights reserved' repo.",
210         "suggestedFix": "",
211         "severity": "low"
212     }
213 ]
214 },

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215     {
216         "doc": "deploy/DEPLOY_ALPHA.md",
217         "overallStatus": "minor-staleness",
218         "summary": "Verified correct except a stale header",
219         "findings": [
220             {
221                 "claim": "header stale deploy live",
222                 "docLocation": "deploy/DEPLOY_ALPHA.md:3",
223                 "verdict": "stale",
224                 "evidence": "droplet/README.md:100 live-verified 2026-06-22",
225                 "suggestedFix": "set header to LIVE",
226                 "severity": "medium"
227             },
228             {
229                 "claim": "host list too narrow",
230                 "docLocation": "deploy/DEPLOY_ALPHA.md:66",
231                 "verdict": "stale",
232                 "evidence": "engine.env.example adds loopback",
233                 "suggestedFix": "add localhost",
234                 "severity": "low"
235             }
236         ]
237     },
238     {
239         "doc": "deploy/droplet/README.md",
240         "overallStatus": "fresh",
241         "summary": "This command-dense droplet quickstart is accurate and matches the
live repo: every cited kit file exists with the described contents, and the engine-behavior
claims (fail-fast boot posture, RALLY_* exposure vars, the Gemini-default-on-CPU gotcha, the
gs:// jail rejection, the --server entryptoint) all verify against the engine repo and
DEPLOY_ALPHA.md. The only items not locally reproducible are the 2026-06-22 droplet live-run
figures (e.g. \"1313 passed\"), which are an off-box assertion ? they are internally plausible
(engine has ~1269 test functions before parametrization) but cannot be confirmed from these
repos.",
242         "findings": [
243             {
244                 "claim": "Four kit files exist: config.json.example, engine.env.example,
khelsutra-engine.service (and ../cloudflared.config.example.yml), each going to the documented
destination with the documented purpose.",
245                 "docLocation": "deploy/droplet/README.md:9-14",
246                 "verdict": "correct",
247                 "evidence": "git ls-files deploy/ lists all four:
deploy/droplet/{config.json.example,engine.env.example,khelsutra-engine.service} and
deploy/cloudflared.config.example.yml; contents read and match (config.json.example has the
security block + cf_team/aud; engine.env.example has the two RALLY_* exposure vars +
RALLY_APP_HOME; the service binds 127.0.0.1:8000; cloudflared.config routes api. ->
127.0.0.1:8000).",
248                 "suggestedFix": "",
249                 "severity": "low"
250             },
251             {
252                 "claim": "The boot fails fast (exit 1, '[STARTUP] refusing to start') if
edge_auth_enabled=true but the jail or cf_team_domain/cf_access_aud is unset.",
253                 "docLocation": "deploy/droplet/README.md:53-54",
254                 "verdict": "correct",
255                 "evidence": "backend/main.py:857 logs '[STARTUP] refusing to start';
backend/config/startup.py:109 EXPOSED_NO_VIDEO_ROOT and :123 EXPOSED_EDGE_AUTH_INCOMPLETE are
both LEVEL_ERROR (startup.py:41,272-276), while EXPOSED_AUTH_EXTRA_MISSING (PyJWT) is LEVEL_WARN
(:135). From engine PR #294 (commit 0c0e064).",
256                 "suggestedFix": "",
257                 "severity": "low"
258             },
259             {
260                 "claim": "indexing.default_segmenter MUST be gemini on a CPU box; the
shipped default yolo_hybrid needs torch/GPU and every job fails 'No module named torch' without

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it.",
261         "docLocation": "deploy/droplet/README.md:109-113",
262         "verdict": "correct",
263         "evidence": "config.json.example:9-11 sets default_segmenter=gemini. Engine
AppConfig default is yolo_hybrid (backend/config/models.py:128), so an all-defaults load on the
droplet (which hand-writes config.json rather than running the --app LIGHT seeder) does land on
yolo_hybrid. The LIGHT 'motion' default (loader.py:227) only applies via
write_light_default_config (indexer --app first-run, PR #310), not this systemd path ? so the
gotcha is accurate for this deploy.",
264         "suggestedFix": "",
265         "severity": "low"
266     },
267     {
268         "claim": "RALLY_ALLOWED_HOSTS is the load-bearing fix; with edge_auth on,
the Host-header guard defaults loopback-only and 400s every tunneled request until set.",
269         "docLocation": "deploy/droplet/README.md:92, engine.env.example:14-17",
270         "verdict": "correct",
271         "evidence": "backend/config/startup.py:143-147 raises
EXPOSED_HOST_ALLOWLIST_LOOPBACK (LEVEL_WARN) when edge_auth is ON and RALLY_ALLOWED_HOSTS is
unset, with the exact 'will 400 forwarded requests' rationale; backend/main.py reads
RALLY_ALLOWED_HOSTS / RALLY_BIND_HOST / RALLY_CORS_ALLOW_ORIGINS. Matches DEPLOY_ALPHA.md §2.2
repro (400->200).",
272         "suggestedFix": "",
273         "severity": "low"
274     },
275     {
276         "claim": "Pasting a gs:// URI is rejected by the jail on the exposed box;
upload is the alpha path.",
277         "docLocation": "deploy/droplet/README.md:96-98",
278         "verdict": "correct",
279         "evidence": "tests/test_api_endpoints.py:608
test_process_hosted_mode_rejects_nonlocal_uri pins the gs:// -> 400 behavior under
allowed_video_root (referenced at :541 as the counterweight); the jail is required to expose
(config.json.example sets allowed_video_root).",
280         "suggestedFix": "",
281         "severity": "low"
282     },
283     {
284         "claim": "Pages build: Root dir web, Build npm run build, Output web/dist,
with VITE_API_BASE = https://api.khelsutra.guru/api.",
285         "docLocation": "deploy/droplet/README.md:73-76",
286         "verdict": "correct",
287         "evidence": "khelsutra web/ SPA exists (git ls-files web/...);
web/package.json build script = 'vite build'; web/src/api/client.ts:25 reads
import.meta.env.VITE_API_BASE (default '/api'), and client.test.ts:32 exercises the exact split
value https://api.khelsutra.guru/api. Vite default outDir 'dist' relative to root web/ =
web/dist.",
288         "suggestedFix": "",
289         "severity": "low"
290     },
291     {
292         "claim": "The shipped engine is run via the auth extra and python -m
backend.main --server, binding 127.0.0.1:8000.",
293         "docLocation": "deploy/droplet/README.md:33,47-49; khelsutra-
engine.service:26",
294         "verdict": "correct",
295         "evidence": "backend/main.py:867 defines the --server flag and :1107 calls
uvicorn.run(app, host=bind_host, port=port); --setup flag at :865; the service ExecStart uses
python -m backend.main --server.",
296         "suggestedFix": "",
297         "severity": "low"
298     },
299     {
300         "claim": "Live-verified on the droplet (2026-06-22): full suite 1313 passed
+ ruff/mypy clean on the box; boot-posture, Host-guard 400->200, path-jail, T4 upload->jail,

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edge-auth 401-gate exercised live; full DNS->Cloudflare->Access->tunnel->engine chain
confirmed.",
301         "docLocation": "deploy/droplet/README.md:100-106",
302         "verdict": "unverifiable",
303         "evidence": "This is an off-box live-run assertion that cannot be reproduced
from the khelsutra or engine repos here. Sanity check: the engine tests/ dir has ~1269 'def
test_' functions (parametrize expansions push the collected/passed count higher), so '1313
passed' is plausible and consistent, not contradicted. Marked unverifiable per the file:line-
evidence discipline; no sign of staleness.",
304         "suggestedFix": "",
305         "severity": "low"
306     },
307     {
308         "claim": "Cost: Cloudflare Pages + Tunnel + Access are free (Access <= 50
users); standing cost ~ $0 + Gemini per run.",
309         "docLocation": "deploy/droplet/README.md:16-18",
310         "verdict": "unverifiable",
311         "evidence": "External Cloudflare pricing claim; not checkable against the
repo. Consistent with DEPLOY_ALPHA.md:3 ('~$0/mo standing ... + Gemini per-run'). No internal
contradiction.",
312         "suggestedFix": "",
313         "severity": "low"
314     }
315 ]
316 },
317 {
318     "doc": "docs/ANNOTATE_AND_TRAIN.md",
319     "overallStatus": "significant-staleness",
320     "summary": "A DRAFT tracker whose marketing-page deliverables (P0-P4 plus the
real annotator screenshots P5 mentions) are all shipped and merged (#15/#22/#23/#25), yet the
doc was last touched 2026-06-20 (#23) and never updated to reflect that. The core
engine/annotator technical claims (schema bridge, ship-gate refusing a commercial ship, MIT
annotator v1.6.4, guardrails) all still verify as correct ? only their file:line citations have
drifted. Main staleness: status still DRAFT despite shipped work, P5 marked Todo though
annotator screenshots are committed, the doc is NOT actually indexed in docs/README.md (claimed
three times), and the page since grew 6-language localization the tracker never anticipated. It
is an archive candidate once honestly closed out.",
321     "findings": [
322     {
323         "claim": "Status: DRAFT ? owner-gated on scope (header, line 3); Last
updated: 2026-06-20.",
324         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:3",
325         "verdict": "stale",
326         "evidence": "git log --follow shows the marketing-page work merged: P0 doc
(#15, 6b62ebf), own-model page P1-P4 (#22, b1110e1), org link (#23, 87520b8). Per $7, P0-P4 are
all marked Done ? yet the header still reads DRAFT. Lifecycle (line 4) is DRAFT->IN
PROGRESS->DONE; with P0-P4 shipped this is at least IN PROGRESS.",
327         "suggestedFix": "Flip Status to IN PROGRESS (only owner-gated P5-P9 remain)
and bump Last updated; or, if the owner considers the marketing surface complete, mark DONE and
archive per the lifecycle note.",
328         "severity": "medium"
329     },
330     {
331         "claim": "P5 'Capture + commit real screenshots ... replace placeholders' ?
Status ? (Todo), Gated Owner.",
332         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:76",
333         "verdict": "stale",
334         "evidence": "PR #25 (1906c82) 'show the real annotator screenshots'
committed site/public/img/annotate-enable.png (133712 bytes) and annotate-tool.png (153133
bytes); own-model.html:111-112 references them as the real VLC-UI shots (no people/footage). The
annotator-UI half of P5 is done and merged, not Todo.",
335         "suggestedFix": "Mark P5 ? for the annotator-UI screenshots (cite #25); if
footage screenshots with consented people are still pending, split that into its own row so the
done part isn't hidden under a blanket Todo.",
336         "severity": "high"

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337         },
338         {
339             "claim": "Doc is 'index[ed] in docs/README.md' (line 5), §7 P0 'indexed in
docs/README.md' (? , #15), and DoD 'doc indexed in docs/README.md'.",
340             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:5,71,91",
341             "verdict": "stale",
342             "evidence": "grep -ni 'annotate|own.model|train.your|ANNOTATE_AND'
docs/README.md returns NOT FOUND; the docs/README.md 'Tracked work' list (lines 1-30) does not
mention ANNOTATE_AND_TRAIN.md or own-model. The claimed index entry does not exist.",
343             "suggestedFix": "Either add an ANNOTATE_AND_TRAIN.md entry to
docs/README.md's 'Tracked work' list, or correct P0/DoD to reflect that it is not yet indexed.",
344             "severity": "high"
345         },
346         {
347             "claim": "Engine ship-gate (gate_verdict, harness.py:98) currently refuses a
commercial ship regardless of F1: WASB provenance C3 unresolved + F1@tIoU not calibrated
(#172).",
348             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:12,31",
349             "verdict": "correct",
350             "evidence": "training/gen0/harness.py:177 returns {'ship': False, ...}; :175
'WASB checkpoint provenance C3 UNRESOLVED ? commercial ship blocked regardless of F1'; :174
'ship-gate F1@tIoU target NOT YET CALIBRATED ... see issue #172'. Fact correct; the cited line
(:98) has drifted ? gate_verdict is now at :142.",
351             "suggestedFix": "Update the line citations: gate_verdict is at
harness.py:142 (not :98).",
352             "severity": "low"
353         },
354         {
355             "claim": "Default synth path: synth_trajectory_from_labels uses
csv.DictReader reading row['start_time']/row['end_time'] (stub_runner.py:139) ? MATCHES the
annotator header exactly.",
356             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:46",
357             "verdict": "correct",
358             "evidence": "backend/pipeline/detectors/stub_runner.py:124 def
synth_trajectory_from_labels; :137 for row in _csv.DictReader(f); :139
rallies.append((float(row['start_time']), float(row['end_time']))). Annotator header
rally_number,start_time,end_time,... confirmed at rally-annotator vlc/rally_annotator.lua:128
and docs/CSV_FORMAT.md:6. Claim accurate.",
359             "suggestedFix": "",
360             "severity": "low"
361         },
362         {
363             "claim": "Canonical GT reader load_gt_intervals accepts start/end OR
start_time/end_time (gt_loader.py:14-90).",
364             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:47",
365             "verdict": "correct",
366             "evidence": "backend/eval/gt_loader.py:30 _START_NAMES = ('start',
'start_time'); :44 def load_gt_intervals; :70 headerless positional col0,col1 path. Both aliases
accepted as claimed.",
367             "suggestedFix": "",
368             "severity": "low"
369         },
370         {
371             "claim": "Bundle attests no_paid_llm_training_labels:True (harness.py:192)
and label source is human (harness.py:178).",
372             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:33,60",
373             "verdict": "correct",
374             "evidence": "training/gen0/harness.py:236 'no_paid_llm_training_labels':
True; :222 'label_source': 'human (rally-annotator golden CSVs)'. Facts correct; cited lines
(:192, :178) have drifted to :236 and :222.",
375             "suggestedFix": "Update line citations: no_paid_llm_training_labels at
harness.py:236, label_source at :222.",
376             "severity": "low"
377         },
378         {

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379         "claim": "cmd_train (__main__.py:231) pulls labels/, joins video<->label by
file MD5 + stem-match labels/<name>.rallies.csv <-> videos/<name>.<ext> (__main__.py:271-280).",
380         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:31,49",
381         "verdict": "correct",
382         "evidence": "backend/worker/__main__.py:332 def cmd_train; :359 lists
{corpus}/labels/; :383-384 fname.replace('.rallies.csv','') stem join. Logic present and
correct; cmd_train line drifted from :231 to :332.",
383         "suggestedFix": "Update cmd_train citation to __main__.py:332.",
384         "severity": "low"
385     },
386     {
387         "claim": "Reuse map: add a /own-model alias in site/server.js:49-51; Worker
env.ASSETS.fetch needs no change (site/src/index.js:84).",
388         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52,73",
389         "verdict": "correct",
390         "evidence": "site/server.js:99 'pathname === \"/own-model\" ? \"/own-
model.html\"' and :41 LOCALIZED_PAGES includes own-model.html (alias shifted off the cited
:49-51). site/src/index.js:84 'return env.ASSETS.fetch(request)' with no /own-model branch ?
confirms the Worker needs no change.",
391         "suggestedFix": "Update the server.js alias citation (now ~:99, not
:49-51).",
392         "severity": "low"
393     },
394     {
395         "claim": "Reuse precedent: screenshots under site/public/img/ (precedent
site/og.png + make_og.py).",
396         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52",
397         "verdict": "correct",
398         "evidence": "git ls-files: site/make_og.py and site/public/og.png both
exist; site/public/img/ holds annotate-enable.png, annotate-tool.png, studio-*.png. The og asset
path is site/public/og.png (the doc wrote 'site/og.png' loosely), make_og.py is at
site/make_og.py.",
399         "suggestedFix": "Minor: the OG image is at site/public/og.png (not
site/og.png).",
400         "severity": "low"
401     },
402     {
403         "claim": "SPA today: web/src/App.tsx is one static screen, no router
(web/package.json); a correction screen needs react-router-dom.",
404         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,64",
405         "verdict": "stale",
406         "evidence": "grep on web/package.json: no react-router dep (claim 'no
router' still TRUE). But App.tsx is no longer 'one static screen' ? it now imports and
conditionally renders SetupWizard (web/src/App.tsx:11,65,242, P2c first-run wizard from PR #88).
The 'one static screen' framing predates the wizard.",
407         "suggestedFix": "Reword to 'no router; App.tsx renders a first-run
SetupWizard + the ingest panel' so the SPA surface description matches the post-#88 code.",
408         "severity": "low"
409     },
410     {
411         "claim": "SPA correction route depends on a new, unbuilt engine endpoint
PATCH /api/segments/{id} (dead columns
candidate_segments.human_label/notes/confirmed_segment_id); P8 ? Gated.",
412         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,79",
413         "verdict": "correct",
414         "evidence": "grep across backend/ for 'api/segments'/'PATCH' on
candidate_segments finds only the candidate_segments table + query_candidate_segments readers
(database.py:217, eval/*.py); no PATCH /api/segments route exists. P8 is still genuinely unbuilt
as stated.",
415         "suggestedFix": "",
416         "severity": "low"
417     },
418     {
419         "claim": "Relation note: peer of LOCAL_APPLIANCE_ARCHITECTURE.md (the latter
lands with PR #13).",

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420         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:6,84",
421         "verdict": "stale",
422         "evidence": "git ls-files confirms docs/LOCAL_APPLIANCE_ARCHITECTURE.md
already exists in the repo; it has long since landed, so the forward-looking 'lands with PR #13'
/ 'cross-link once PR #13 merges' phrasing is outdated.",
423         "suggestedFix": "Drop the 'lands with PR #13' future-tense framing ? the doc
exists now; add the cross-link.",
424         "severity": "low"
425     },
426     {
427         "claim": "rally-annotator is a single-file VLC Lua extension, VERSION 1.6.4,
MIT-licensed, public at github.com/Khelsutra/rally-annotator.",
428         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:29",
429         "verdict": "correct",
430         "evidence": "rally-annotator/vlc/rally_annotator.lua:1 '(v1.6.4)'; LICENSE
first line 'MIT License'; own-model.html:109/191 links to https://github.com/Khelsutra/rally-
annotator (org-link migration was PR #23, captured in the doc's own history).",
431         "suggestedFix": "",
432         "severity": "low"
433     },
434     {
435         "claim": "$7 tracker: P1/P2/P3/P4 all ? via '(this PR)'.",
436         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:72-75",
437         "verdict": "correct",
438         "evidence": "own-model.html exists (P1); /own-model alias in server.js:99
(P2); nav cross-link index.html:255 + footer :620 (P3); verbatim annotator content + GitHub link
in own-model.html:108-191 (P4). All shipped ? git history shows these in PR #22 (b1110e1,
'annotate-train P1-P4'), so '(this PR)' should now read '#22'.",
439         "suggestedFix": "Replace the '(this PR)' placeholders in the PR column with
#22 (and #23 for the org-link follow-up) so the tracker cites real merged PRs.",
440         "severity": "medium"
441     }
442 ]
443 },
444 {
445     "doc": "docs/DRY_RUN_CUJ.md",
446     "overallStatus": "minor-staleness",
447     "summary": "The doc's substantive engineering claims are accurate and well-
grounded: the SPA back-half + B-P2 front-half ingest screen exist
(web/src/components/IngestPanel.tsx, web/src/hooks/useIngestJob.ts, #56), the engine bucket-
ingest routing is verified in code, and the three cited engine PRs (#280/#281/#289) plus all
four cited engine tests exist. The staleness is in metadata/framing: the header still says
Status: DRAFT though docs/README.md correctly calls it DELIVERED with DoD $6 boxes 1-3 done, and
the doc repeatedly describes the multilingual workflow as \"en/kn/hi\" (3 langs) when the SPA
now supports 6 (es/da/id added in #87). It is a completed tracker and an archive candidate once
the parked droplet box is closed.",
448     "findings": [
449         {
450             "claim": "Status: DRAFT ? owner-gated . ... Goal for this session.",
451             "docLocation": "docs/DRY_RUN_CUJ.md:3",
452             "verdict": "stale",
453             "evidence": "docs/README.md:4 marks this doc '? DELIVERED (2026-06-21)'; DoD
$6 boxes 1-3 are all checked [x] (DRY_RUN_CUJ.md:47-49) and the work shipped (#53/#55/#56/#57).
A DRAFT/'goal for this session' header no longer matches the completed, delivered state.",
454             "suggestedFix": "Flip the header to Status: DONE / DELIVERED (2026-06-21)
and drop 'Goal for this session'; note the only open box (droplet, $6 box 4) is owner-parked.
Consider archiving once that box closes.",
455             "severity": "medium"
456         },
457         {
458             "claim": "The SPA is 'Fully en/kn/hi (the switcher / ?lang=)' and the
multilingual workflow is en/kn/hi.",
459             "docLocation": "docs/DRY_RUN_CUJ.md:11",
460             "verdict": "stale",
461             "evidence": "web/src/i18n/index.ts:25 ? `export const SUPPORTED_LOCALES =

```

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[\"en\\\", \"kn\\\", \"hi\\\", \"es\\\", \"da\\\", \"id\\\"]`; es/da/id were added to the SPA in PR #87
(cf59fbf 'add es/da/id to the marketing site (parity with the SPA's 6 languages)'). The SPA now
supports 6 languages, not 3.",
462     "suggestedFix": "Update 'en/kn/hi' to 'en/kn/hi/es/da/id (6 languages)'
where it describes the current switcher; the Hindi/Kannada run results in §7 stay accurate as
historical record.",
463     "severity": "low"
464 },
465 {
466     "claim": "This SPA front-half (ingest/process UI) is the *only* net-new work
for Path B (= B-P2); the SPA has no ingest/process UI today.",
467     "docLocation": "docs/DRY_RUN_CUJ.md:14",
468     "verdict": "correct",
469     "evidence": "B-P2 was net-new work at draft time and has since shipped:
web/src/components/IngestPanel.tsx + web/src/hooks/useIngestJob.ts (first landed in PR #56
'feat(web): B-P2 ? in-UI bucket ingest', 4525c5d, 2026-06-21). useIngestJob.ts drives POST
/api/process (processVideo) ? poll GET /api/jobs/{id} (getJob) ? hand video_id to App, exactly
as scoped.",
470     "suggestedFix": "",
471     "severity": "low"
472 },
473 {
474     "claim": "POST /api/process accepts gs://gcs://s3://local:// URIs and skips
the local-existence check for non-local backends; resolve via storage.resolve_input_ref;
gs://?gcs normalization.",
475     "docLocation": "docs/DRY_RUN_CUJ.md:15,68-72",
476     "verdict": "correct",
477     "evidence": "Engine backend/main.py:581 `input_ref =
storage.resolve_input_ref(req.video_path, ...)` and main.py:590 `if input_ref.backend ==
'local' and not os.path.exists(input_ref.key):` (the non-local skip), with the fetch at
main.py:458 `storage.acquire_local_input(...)`; backend/storage/ref.py:60 `if scheme == 'gs':
scheme = 'gcs'`. Substance matches; the exact line numbers 159/274/283 cited in the doc have
drifted as the engine shipped further PRs.",
478     "suggestedFix": "Optionally refresh the cited line numbers (now
~main.py:581/590/458, ref.py:60) since the engine surface moved; the routing logic itself is
intact.",
479     "severity": "low"
480 },
481 {
482     "claim": "Tested by
tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches,
::test_process_hosted_mode_rejects_nonlocal_uri, ::test_process_unknown_scheme_returns_400,
::test_cli_nonlocal_input_fetches_and_keeps_uri.",
483     "docLocation": "docs/DRY_RUN_CUJ.md:15,76",
484     "verdict": "correct",
485     "evidence": "Engine repo grep: tests/test_api_endpoints.py:508
test_process_nonlocal_uri_skips_exists_and_fetches; :577
test_process_unknown_scheme_returns_400; :608 test_process_hosted_mode_rejects_nonlocal_uri;
:627 test_cli_nonlocal_input_fetches_and_keeps_uri. All four exist with the asserted names.",
486     "suggestedFix": "",
487     "severity": "low"
488 },
489 {
490     "claim": "B-P1 (gs://?/api/process?API-DB video_id) already shipped in
engine M2 / #280; the job-drain hang fixed in #281; the compile-from-bucket 500 fixed in #289.",
491     "docLocation": "docs/DRY_RUN_CUJ.md:15,63,68,88",
492     "verdict": "correct",
493     "evidence": "Engine git log: e111add 'feat(serving): M2 ? URI-aware input
... (#280)'; 41d45fd 'fix(server): drain async jobs under `python -m backend.main` (double-
module hang) (#281)'; e68b583 'fix(api): resolve a bucket-ingested video's source to a local
file before stitching (#289)'. All three PRs exist with descriptions matching the doc.",
494     "suggestedFix": "",
495     "severity": "low"
496 },
497 {

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498         "claim": "DoD §6 box 3: in-UI Path-B bucket?reel CUJ DONE (live e2e
2026-06-21) ? 12 rallies loaded ? reel downloaded; box 4 (CPU droplet) still open/un-
provisioned.",
499         "docLocation": "docs/DRY_RUN_CUJ.md:49-50",
500         "verdict": "correct",
501         "evidence": "Box 3 is corroborated by PR #57 'docs(dry-run): Path-B live e2e
DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3 ?)' and the merged B-P2 SPA (#56).
docs/README.md:6-7 confirms box 4 (droplet, option 3) remains 'parked by the owner (GPU
subscription)'. The run-result specifics (12 rallies, reel sizes, GCS screencast paths) are
runtime artifacts not locally reproducible.",
502         "suggestedFix": "",
503         "severity": "low"
504     },
505     {
506         "claim": "Run-result specifics: 12 rallies / video_id 05e0e577 /
50.9MB+76.5s reels / screencasts at gs://khelsutra-rally-corpus/promo/*.mp4 and the staged
bucket video gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4.",
507         "docLocation": "docs/DRY_RUN_CUJ.md:54-60,82-85",
508         "verdict": "unverifiable",
509         "evidence": "These are GCS bucket objects and a one-off local run's outputs;
they cannot be checked from the repo. No counter-evidence found ? the surrounding code/PR claims
that frame them are all verified.",
510         "suggestedFix": "",
511         "severity": "low"
512     }
513 ]
514 },
515 {
516     "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
517     "overallStatus": "significant-staleness",
518     "summary": "A still-active DRAFT tracker (last updated 2026-06-21) whose §9
progress and several \"Absent (net-new)\" engine-fact anchors have drifted: the engine has since
shipped /api/capabilities, exposure-safety boot posture + an edge-auth layer, and merged
native_wasb_runner to master, while the alpha deploy went LIVE on 2026-06-22
(app.khelsutra.guru). The khelsutra-side claims (PR refs #11/#56/#64/#66, IngestPanel,
VITE_API_BASE) check out; the staleness is concentrated in engine status and the deploy/DoD
progress. Not yet an archive candidate (open owner gates remain) but needs a refresh pass.",
519     "findings": [
520     {
521         "claim": "§3 lists `GET /api/capabilities` under 'Absent (net-new): ...
`[verified ? grep returns none]`; §4.6 Reuse map lists it under 'Net-new'; §9 T7 (capabilities
+ health + banner) is ? Todo.",
522         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:49,113,164",
523         "verdict": "stale",
524         "evidence": "engine origin/master backend/main.py:340
`@app.get(\"/api/capabilities\")` (and main.py:307 `/api/health`, main.py:334 `/api/version`)
all exist on master. git grep '/api/capabilities' origin/master returns a hit.",
525         "suggestedFix": "Move `/api/capabilities` out of 'Absent (net-new)' and out
of §4.6 'Net-new'; mark the engine half of T7 (capabilities+health) as shipped (?/?), leaving
only the SPA banner pending.",
526         "severity": "high"
527     },
528     {
529         "claim": "TL;DR + Risk table: 'the engine has zero app-layer auth and CORS
`*` `[verified backend/main.py:42]` ? structural safety only because the engine binds
127.0.0.1.",
530         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:14,30,103,128",
531         "verdict": "stale",
532         "evidence": "engine origin/master backend/main.py:82 uses
`allow_origins=cors_origins_from_env()` (default ['*'] but env-lockable) with
`allow_credentials=False`; main.py:109 adds TrustedHostMiddleware (DNS-rebinding/Host
allowlist); backend/api/auth.py + config/startup.py enforce_startup_checks + M9 `Cf-Access-
Authenticated-User-Email` `owner_id` now exist on master. CORS is no longer a hardcoded bare
`*`, and an app-layer edge-auth integration now exists.",
533         "suggestedFix": "Reword to: CORS default is `*` but now env-configurable

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(allow_credentials off), and the engine now ships a Host-header guard + an edge-auth integration
(Cf-Access trust) + exposure-safety boot posture ? so 'zero app-layer auth' is no longer
literally true. Re-anchor the line (main.py:42 no longer points at CORS).",
534         "severity": "high"
535     },
536     {
537         "claim": "$3 / $4.3: 'Native WASB is in-flight on branch `feat/native-wasb-
runner`: `native_wasb_runner.py` exists `[verified]`'.",
538         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:45,96",
539         "verdict": "stale",
540         "evidence": "engine origin/master now contains
backend/pipeline/detectors/native_wasb_runner.py and tests/test_native_wasb_runner.py (git ls-
tree origin/master). The runner has merged to master; the `feat/native-wasb-runner` branch still
exists but the file is no longer branch-only. (trajectory_hybrid._build_native_runner still
raises NotImplementedError at trajectory_hybrid.py:61 ? that sub-claim is still correct.)",
541         "suggestedFix": "Change 'in-flight on branch feat/native-wasb-runner' to
'native_wasb_runner.py is on master; trajectory_hybrid still refuses, fusion builds it; owner
byte-parity gate still pending.'",
542         "severity": "medium"
543     },
544     {
545         "claim": "$9 alpha-deploy block: 'the actual stand-up (tunnel + Pages + CF
Access + key rotation) is the **owner** step'; $11 DoD box 1 (CUJ-1 e2e on droplet behind gate)
and box 2 (raw :8000 refused, tested) unchecked.",
546         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:175,182,183",
547         "verdict": "stale",
548         "evidence": "deploy/droplet/README.md:100-106 '? Live-verified on the
droplet (2026-06-22)': engine booted on 168.144.126.176; suite 1313 passed; boot-posture fail-
fast, Host-guard 400?200, path-jail, T4 upload?jail, and edge-auth 401-gate all exercised live;
full DNS?Cloudflare?Access?tunnel?engine chain confirmed; app.khelsutra.guru up. Only Gemini-key
rotation + a browser OTP login remain.",
549         "suggestedFix": "Update $9 to note the alpha stand-up is LIVE (2026-06-22,
app.khelsutra.guru, posture test exercised) with only key rotation + browser-OTP login
outstanding; tick/qualify the corresponding DoD boxes.",
550         "severity": "high"
551     },
552     {
553         "claim": "Status header: 'Status: DRAFT ... Last updated: 2026-06-21' with
$9 the source of truth.",
554         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:3",
555         "verdict": "stale",
556         "evidence": "khelsutra git log shows merges through #94 (master HEAD
50b2bd7) and the alpha went live 2026-06-22, both after the doc's 2026-06-21 'last updated'.
Multiple $9 rows and engine facts have moved since (T7 engine half, alpha deploy, native_wasb on
master).",
557         "suggestedFix": "Bump 'Last updated' and consider promoting DRAFT?IN
PROGRESS; refresh $9 rows to current reality.",
558         "severity": "medium"
559     },
560     {
561         "claim": "D4 'Embed RallyPicker ... PR [#11]' and $9 T4 '[#66]', T12 'SPA
#56', alpha-deploy '[#64]' / 'VITE_API_BASE'.",
562         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:33,161,169,175",
563         "verdict": "correct",
564         "evidence": "khelsutra git log: cb74ce5 '...RallyPicker (#11)'; ca619fb 'T4
? in-UI file upload (#66)'; 4525c5d 'B-P2 ? in-UI bucket ingest (#56)'; 8bbc557 'VITE_API_BASE
(#64)'. web/src/components/IngestPanel.tsx handles both chunked upload (T4) and a gs:// URI
(T12); web/src/api/client.ts:25 reads VITE_API_BASE.",
565         "suggestedFix": "",
566         "severity": "low"
567     },
568     {
569         "claim": "T12 (bucket-URI ingest) marked ? : 'engine accepts URIs (M2/#280) +
SPA ingest screen shipped (B-P2, #56); in-UI gs:///reel CUJ proven live e2e (DRY_RUN_CUJ.md
$9)'.",

```

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570         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:169",
571         "verdict": "correct",
572         "evidence": "docs/DRY_RUN_CUJ.md:80-92 §9 records the live in-UI gs://?reel
CUJ (12 rallies ? reel downloaded) and the compile-from-bucket 500 fix (engine #289);
IngestPanel.tsx accepts a gs:// URI. khelsutra #56 (B-P2) is merged.",
573         "suggestedFix": "",
574         "severity": "low"
575     },
576     {
577         "claim": "T11 retention marked ?: sweep tool shipped (engine #299), auto-
scheduling tracked (engine #301).",
578         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:168,130",
579         "verdict": "unverifiable",
580         "evidence": "These reference the engine repo's GitHub PR #299 / issue #301
(Khelsutra/badminton-highlight-indexer). Not falsifiable from the local khelsutra checkout; the
local engine checkout is on a feature branch and I did not confirm the specific sweep tool by PR
number. The khelsutra-side row metadata is internally consistent (#69 flipped T11 to ? per git
log bea8a2a).",
581         "suggestedFix": "",
582         "severity": "low"
583     },
584     {
585         "claim": "§3 Droplet reality: co-hosts the marketing site at
`/srv/khelsutra-site`, node `:8787`, engine at `~/rally`; MinIO 9000/9001 + GCS emulator 9023.",
586         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48",
587         "verdict": "unverifiable",
588         "evidence": "These are live-droplet filesystem/port facts (168.144.126.176)
not checkable from the local repo. deploy/DEPLOY_ALPHA.md:31 and deploy/droplet/README.md:4
corroborate the IP/CPU-no-GPU/blr1 framing, but the exact paths/ports (/srv/khelsutra-site,
:8787, ~/rally) are not in-repo verifiable.",
589         "suggestedFix": "",
590         "severity": "low"
591     },

```

4. user (2026-06-22T09:46:29.266Z)

```

1      # Interactive per-rally selection ? reel builder (RallyPicker)
2
3      > **Status:** `IN PROGRESS` ? **MVP / M0 (P1?P5) COMPLETE 2026-06-20** (pick ? query ?
select ? compile ? download, on existing engine endpoints). **Validated live e2e against a REAL
engine (Gemini) via the in-UI bucket?reel CUJ ? `DRY_RUN_CUJ.md` §9, 2026-06-21** (supersedes
the original stub-engine verification). Design **APPROVED by owner 2026-06-20**. Downstream
milestones M1?M5 (multi-video auth, per-rally preview endpoint, corpus loop, localized NL)
remain owner-gated. . **Owner:** Avi . **Created:** 2026-06-20 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6      > **Relation:** second major tracked work item in the `khelsutra` repo (peer of
`REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
7      > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine
code this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9      ---
10
11     ## 0. TL;DR
12     Build the **pick-rallies ? reel** surface as a **hybrid**: a selectable rally
**list/grid** is the workhorse; a **deterministic query bar** ("rallies over 10s, longest
first") pre-selects into the **same** selection set; a sticky footer compiles the selection into
one downloadable MP4. The entire MVP runs on **existing engine endpoints** ? `POST /api/query` ?

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`POST /api/compile` ? `GET /api/highlights` `[verified backend/main.py:331-340, 342-396,
307-328]` ? with zero new engine intelligence and zero new endpoints. The single most
important caveat: queries and chips are scoped to length / count / order / shot-count only;
shot-type, player, and score are roadmap-only (no detector, no column) and must not be
faked. The biggest UX-vs-effort tension: no per-rally preview is possible today ? only the
final compiled reel can stream ? so MVP selects on text metadata, and visual preview/timeline
is a deliberate later milestone.
13
14     ## 1. Problem & goal
15     Today the alpha can detect rallies and stitch a reel, but reel-building is either a
silent `stitching.min_shot_count` threshold `[verified backend/main.py:362-383]` or a static
checkbox list whose data plumbing is even broken ? the bundled static UI reads `data.rallies`
while the API returns `play_segments`, so it renders nothing `[verified
backend/api/static/app.js:344 vs backend/main.py:340]`. The owner's framing is dynamic per-
video selection, not a fixed threshold: a coach should say "give me the long rallies from this
session" and get exactly those, then fine-tune by hand.
16
17     The concrete outcome ("good"): A coach uploads a 40-minute session; the engine finds
60 rallies. They type "top 15 longest", eyeball the auto-selected cards, untick two warm-up
rallies, name it "Tuesday ? long points", and download a ~6-minute reel ? without learning any
threshold knob.
18
19     Goal: the smallest honest surface that turns engine-detected rallies into a user-
curated, downloadable reel, with both fast (query) and precise (checkbox) selection
sharing one state.
20
21     Non-goals (alpha): no conversational LLM (`UI_PROPOSAL.md:66` defers it); no shot-
type/player/score filters (no data exists); no per-rally video preview while it requires unbuilt
streaming; no multi-tenant auth (single-box alpha).
22
23     ## 2. Decisions locked
24     | # | Decision | Source / date | Implication |
25     |---|-----|-----|-----|
26     | D1 | Dominant surface = selectable rally list/grid | Synthesis 2026-06-20 (Grid
scored 15, fit-to-data 4) | Scales to long matches via bulk+filter; shows honest fields only |
27     | D2 | Fast entry = deterministic client-side `parseQuery()` (no LLM) | `[verified
UI_PROPOSAL.md:66]` conversational QA deferred | Predictable, unit-testable, offline, zero API
cost |
28     | D3 | One shared `Set<segment_id>`, mutated by query + checkbox + (future) timeline
| Synthesis 2026-06-20 | No second compile path can diverge from what the user sees |
29     | D4 | Compile via existing `POST /api/compile {video_id, segment_ids, output_filename}`
| `[verified backend/main.py:342-396]` | Selection order is harmless ? stitcher
`merge_overlapping` de-dupes `[verified compiler.py:80-105, 311-317]` |
30     | D5 | Query grammar (alpha) = length, count, order, shot_count only | `[verified
database.py:122-135, gemini.py:470-485]` | Only fields the payload actually returns; grey out
the rest |
31     | D6 | MVP backend = zero new endpoints | `[verified]` reuse map (§4) | Ships the
full loop on the already-shipped contract |
32     | D7 | Replace the silent `min_shot_count` cut with explicit user selection | Owner
framing + `[verified main.py:362-383]` | UI warns about the floor; it never hides why a rally
vanished |
33     | D8 | Default selection = ON (curate-by-removal) | Owner 2026-06-20 | Rallies load
all-selected (matches today's "all rallies" reel); user removes warm-ups. Footer/empty UX
assumes a non-empty start |
34     | D9 | Preview-before-select is NOT a ship-blocker for the MVP | Owner 2026-06-20 |
MVP selects on text metadata + previews the final compiled reel; the source/clip-stream
endpoint (per-rally preview) is the next milestone (P9), not MVP |
35     | D10 | MVP is job-tethered (carry `result.video_id`); no `GET /api/videos` | Owner
2026-06-20 | Smallest honest slice; multi-video home + identity scoping is P8 (gated) |
36     | D11 | `min_shot_count`: warn, do NOT override in the MVP | Owner 2026-06-20 | The
UI surfaces the floor; an explicit-selection override of the compile default is deferred (would
trigger the Launch Report convention) |
37
38     ## 3. Foundation ? what the engine gives us today `[verified this session]`
39     The whole design rests on the engine's real, already-shipped contract. Read against the

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engine repo (`badminton-highlight-indexer`):
40
41     - **The rally store is `play_segments`** ? columns `id, video_id, start_time, end_time,
duration, server, receiver, metadata` `[verified database.py:124-135]`. `id` is the integer the
UI selects and passes to compile; `start_time/end_time/duration` are REAL seconds (duration
computed at insert).
42     - **Listing rallies = `POST /api/query {video_id}` ? `{play_segments:[...]}`** ? the
*only* rally-list path today; it JSON-parses `metadata` and joins `events` `[verified
main.py:331-340, database.py:427-475]`.
43     - **Building a reel = `POST /api/compile {video_id, segment_ids:[int],
output_filename}`** ? loads `videos.filepath`, filters all segments to the requested
`segment_ids`, applies the `min_shot_count` floor, extracts `time_pairs=[(start,end)]`, and
calls `VideoCompiler.compile_highlights(raw_path, segments, name)` `[verified main.py:342-396,
compiler.py:303]`, which **merges overlaps** then per-clip re-encodes ? concat ? one MP4
`[verified compiler.py:311-317]`. Selection order/duplicates are therefore safe.
44     - **Downloading = `GET /api/highlights/{video_id}/{filename}`** ? streams the compiled
MP4. Compile returns a **server-local filepath, NOT a URL** `[verified main.py:394]`, so the SPA
constructs this download URL itself. The compiled reel is the **only streamable video that
exists today** `[verified main.py:307-328]`.
45     - **The floor knob is readable** ? `GET /api/config` exposes `stitching.min_shot_count`
`[verified main.py:121-123]`, so the UI can warn before a selected rally is silently dropped.
46     - **Job entry** ? `GET /api/jobs/{job_id}` returns `result.video_id` `[verified
main.py:274-293]`, which is how the MVP reaches a video without a (nonexistent) list-videos
endpoint.
47
48     **Honest fields vs fabricated/roadmap (critical for the UI):**
49     - **Honest today** (default `default_segmenter: "gemini"`, per `config.json`
`[verified]`): `start_time`, `end_time`, `duration`, and `metadata.shot_count` (Gemini's
`shuttle_exchange_count`, fallback `max(3, int(dur/0.8))`) + `metadata.shot_count_confidence` (a
**string**, often `"Unknown"`) `[verified gemini.py:470-485]`. On the gemini path
`server`/`receiver` are written `None` ("no fabricated data") `[verified gemini.py:480-487]`.
50     - **Fabricated ? do NOT surface:** the `motion` demo segmenter fills `server`/`receiver`
via `random.sample(...)` , a random `shot_count`, and random `events` (serve/smash/foul/winner)
`[verified motion.py:191-198]`. `query` returns these columns + `events`, but they are not
ground truth and must stay hidden.
51     - **Roadmap ? no column, no detector:** shot-type per shot, player identity, match/point
score, game/set grouping. None exist anywhere in the schema; do not consume them `[verified
database.py whole-schema 122-148]`.
52     - **Correction-loop columns exist but are dead:**
`candidate_segments.human_label/notes/confirmed_segment_id` are defined but no endpoint writes
them `[verified database.py:166-168]` ? that's the M4 corpus extension (engine PR-4/B6,
`UI_ALPHA_EXECUTION_PLAN.md`).
53
54     ## 4. Design / architecture
55
56     **Reuse map (existing ? `[verified]`, no change for MVP):** `POST /api/query` (list) ?
`POST /api/compile` (stitch) ? `GET /api/highlights/{video_id}/{filename}` (download); `GET
/api/config` (floor) ; `GET /api/jobs/{id}` (carry `result.video_id`). **All net-new code is the
SPA** in `kheisutra` `web/` (Vite + React + TS + Tailwind, per `web/README.md`), plus a small
set of *optional* real-gap engine enablers ($4.3) ? none required for MVP.
57
58     **4.1 Components (net-new SPA)**
59     - **QueryBar + `parseQuery()`** ? a *pure, deterministic, unit-tested* string ?
`{filter, sort}` compiler over `duration` / `ordinal` / `shot_count`. Renders a **parsed-pill**
("filter: duration > 10s · sort: duration desc") so the interpretation is transparent and hand-
correctable; greys out shot-type/player/score tokens with an inline "not available yet". No LLM,
no network call.
60     - **RallyList/Grid** ? one card per `play_segment`: ordinal (by `start_time`), `mm:ss`
start, duration (`end?start`), and a shot-count + confidence chip **only when
`metadata.shot_count` is present** (absent on detector paths that don't write it). Checkbox per
card; the thumbnail area is a placeholder (no endpoint yet).
61     - **`useRallySelection()` hook** ? owns `selectedIds:Set<number>`, `displayOrder`,
`apply(queryResult)`, `toggle(id)`, and bulk Select-all / Clear / Invert. Unit-tested (matches
the exec-plan's "API client + review-state logic" bar).
62     - **SelectionFooter (sticky)** ? live "N clips / total M:SS"; a **`min_shot_count`

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warning** when a selected rally would be dropped at compile (read from `GET /api/config`); reel-
name input; one **Build** button.
63   - **CompileResult** ? `<video src=GET /api/highlights/...>` preview of the *compiled
reel* + a download link.
64   - **TypedApiClient** ? `query`, `compile`, `buildDownloadUrl`; the typed seam to the
engine.
65
66   **4.2 Data flow** `[verified anchors inline]`
67   Gemini segmenter (process time) writes `play_segments {id, start/end_time, duration,
metadata.shot_count/confidence/detector}` `[gemini.py:461-498]`
68   ? SPA `POST /api/query {video_id}` ? `{play_segments:[...]}` `[main.py:331-340]`
(**fix** the static UI's `data.rallies` read ? `play_segments` `[app.js:344]`)
69   ? normalize to `RallyCard[]` + one shared `selectedIds:Set`;
`server`/`receiver`/`events` are present in the payload but motion-only fabrications ? **not
surfaced** `[motion.py:191-198]`
70   ? all input modes mutate the **same** `selectedIds`: (a) `parseQuery()` predicate over
duration/ordinal/shot_count; (b) checkbox toggle + bulk ops; (c) *(post-MVP)* timeline band
71   ? footer recomputes count + ?duration **client-side**, and warns for any selected rally
with `shot_count < stitching.min_shot_count` `[main.py:362-383]`
72   ? **Build** ? `POST /api/compile {video_id, segment_ids: [...selectedIds],
output_filename}` ? stitcher de-dupes via `_merge_overlapping` `[compiler.py:311-317]`, cuts
`time_pairs` from `videos.filepath`, re-encodes per clip ? concat ? one MP4 `[main.py:386-393,
compiler.py:303]`
73   ? compile returns a **server-local filepath, not a URL** `[main.py:394]` ? SPA builds
`GET /api/highlights/{video_id}/{output_filename}` to preview in `<video>` + download
`[main.py:307-328]`.
74
75   **4.3 Engine API additions (none for MVP; tagged against real gaps)**
76   | Endpoint | Purpose | Status |
77   |---|---|---|
78   | `POST /api/query` | List rallies (only path) ? fix SPA to read `play_segments` |
**exists** |
79   | `POST /api/compile` | Stitch selected `segment_ids` ? MP4 | **exists** |
80   | `GET /api/highlights/{video_id}/{filename}` | Stream/download compiled reel |
**exists** |
81   | `GET /api/config` | Read `stitching.min_shot_count` for the warning | **exists** |
82   | `GET /api/jobs/{job_id}` | Carry `result.video_id` into the picker | **exists** |
83   | `POST /api/compile` ? add `download_url` | Return `/api/highlights/...` so SPA stops
hand-building it (additive, low-risk; today returns only a filepath `main.py:394`) | **modify**
|
84   | `GET /api/health` | Engine-offline banner ping the SPA is specified to use
(`HOSTING_PLAN.md:42`) | **new** |
85   | `GET /api/videos/{video_id}/rallies` | RESTful/cacheable alias returning the same
`{play_segments}` shape | **new** |
86   | `GET /api/videos` | List a user's videos for a Match picker (only `get_video(id)`
exists, `database.py:418`) ? needs DB enumerate + identity scoping | **new** |
87   | `GET /api/videos/{video_id}/source` (Range) *or* `/clip?start=&end=` | Stream
source/per-span video so a rally previews before selection ? neither exists | **new** |
88   | `GET /api/videos/{video_id}/rallies/{segment_id}/thumbnail` | Per-rally poster still
(no thumbnail column; WASB frames are transient) | **new** |
89   | `PATCH /api/segments/{id}` | Persist accept/reject/nudge ?
`candidate_segments.human_label/notes/...` (columns exist, unwritten ? corpus loop) | **new** |
90
91   **? Dead-ends / do-NOT:** never surface `server`/`receiver`/`events` (motion-only
randoms `[motion.py:191-198]`); never consume shot-type/player/score (no data); never render a
confidence **meter** ? `shot_count_confidence` is a Gemini *string* (often `Unknown`) and
`play_segments` has **no** numeric confidence column.
92
93   **4.4 Observability (first-class, per `[memory: feedback-launch-quality-bar]`).** Every
P3 user action emits a **structured JSON log line** (`web/src/lib/log.ts`) so a journey is
traceable from text to selection ? the SPA-side "runlog". Each query application generates **one
`cid`** shared by every line of that journey; the future `api/compile` call (P5) reuses the
same `cid` to thread the trace into the engine. Events (all on the P3 path):
94
95   | event | level | when | key fields |

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96      |---|---|---|---|
97      | `query.parse` | info | every apply attempt | `cid`, `source` (typed/example),
`recognized`, `filters`, `sort`, `limit`, `unavailable` ? the exact interpretation |
98      | `query.unavailable_request` | info | the query names a shot-type/player/score | `cid`,
`raw`, `unavailable[]` ? **demand signal** for roadmap features (what users ask for that we
can't yet do) |
99      | `query.skip` | info | nothing actionable parsed | `cid`, `raw`, `reason` |
100     | `query.apply` | info | a recognized query is applied | `cid`, counts
(`total`/`selected`), `selected_ids`, `dropped_unavailable` |
101     | `query.empty_result` | **warn** | recognized query matches **zero** rallies | `cid` +
full interpretation ? a looks-empty grid is **never silent** |
102     | `query.apply_failed` | **error** | the (pure) parser/selector unexpectedly throws |
`cid`, `raw`, `error` ? loud, never swallowed |
103     | `selection.bulk` | info | Select-all / Clear / Invert | `cid`, `op`, `result_count` |
104     | `rally.toggle` | debug | a card is added/removed | `id`, `action` |
105
106     No silent drops: unsupported concepts (`query.unavailable_request` +
`dropped_unavailable`) and empty results (`query.empty_result` warn) are always logged. Verified
live in-browser (events fire with correct `cid`/fields, zero console errors).
107
108     ## 5. Risk table
109     | Risk | Severity | Mitigation |
110     |---|---|---|
111     | **No source/clip/frame/thumbnail endpoint** ? only compiled reels stream
`[main.py:307-328]` | High (UX) | MVP selects on text metadata + previews the *final* reel;
preview/timeline deferred to M3 behind a new endpoint |
112     | `min_shot_count` silently drops selected rallies `[main.py:362-383]` | High (looks
broken) | Footer warns *before* Build; D7 makes selection explicit; an override is an owner gate
(Launch-Report-worthy `[memory: launch-report-convention]`) |
113     | No match-duration field ? a timeline must derive `total = max(end_time)`, mis-
rendering pre/post dead air | Med | Timeline deferred to M3; never over-promise it as a true
scrubber until source streams |
114     | `shot_count` is an *estimate* (`shuttle_exchange_count` w/ `max(3,int(dur/0.8))`
fallback) `[gemin.py:470-485]` | Med | Label "approx."; confidence shown as the raw string, not
a meter |
115     | No auth / user scoping anywhere (CORS `*`, `videos.user_id` never set/filtered) | High
for multi-tester | Fine for single-box local alpha; a ship-blocker for any shared host ? gate
`GET /api/videos` behind PR-3 scoping |
116     | compile returns a filepath not a URL `[main.py:394]` | Low | Client builds the URL for
MVP; `download_url` field (M1) removes the brittleness |
117     | Surface sprawl (query + grid + future timeline + future correction) | Med | Deliberate
IA: query+grid = one screen; correction = separate ? corpus screen (owner call, $8) |
118
119     ## 6. Honest limits ? what this does NOT do
120     A green SPA build + passing `parseQuery`/selection unit tests prove the **picker logic
only** ? **not** that the engine's rally boundaries are correct (the measured **rally-END**
weakness is a separate quality track, unaffected by this UI). "Conversational" here means a
**deterministic four-dimension parser**, not natural-language understanding. Nothing is
implemented in `web/` yet ? this is `[design]`. The feature curates *what the engine already
found*; it cannot surface a rally the detector missed, nor split/merge a mis-bounded one (that's
the M4 correction loop).
121
122     ## 7. Deliverables & progress tracker ? **source of truth**
123     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. CI (`npm test`) must stay green; merge to `master`
auto-deploys the site (docs/`web/` changes don't touch the live `site/` Worker).
124
125     | ID | Deliverable | Depends on | Gated? | Status | PR |
126     |---|-----|-----|-----|-----|---|
127     | P0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | No | ? | |
[#11](https://github.com/avidullu/khelsutra/pull/11) |
128     | P1 | SPA scaffold (Vite+React+TS+Tailwind) in `web/` + typed API client; dev proxy to
engine `:8000` | P0 | No | ? | [#14](https://github.com/avidullu/khelsutra/pull/14) |
129     | P2 | RallyList/Grid (honest fields only) + `useRallySelection()` hook (unit-tested) |
P1 | No | ? | [#21](https://github.com/avidullu/khelsutra/pull/21) |

```

```

130 | P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable
tokens) | P2 | No | ? | (this PR) |
131 | P4 | SelectionFooter + `min_shot_count` warning (from `GET /api/config`) | P2 | No | ?
| (this PR) |
132 | P5 | Build ? `POST /api/compile` ? client-built download URL + `` preview of
the reel | P2 | No | ? | (this PR) |
133 | P6 | *(engine repo)* `GET /api/health` + offline banner; `/api/compile` returns
`download_url` | ? | No | ? | ? |
134 | P7 | *(engine repo)* `GET /api/videos/{id}/rallies` RESTful alias | ? | No | ? | ? |
135 | P8 | *(engine repo)* `GET /api/videos` + PR-3 identity scoping ? A1 Matches screen |
P7 | ? owner | ? | ? |
136 | P9 | *(engine repo)* source/clip stream + timeline + per-rally preview/thumbnail | P8
| ? owner | ? | ? |
137 | P10 | *(engine repo)* `PATCH /api/segments/{id}` corpus loop (PR-4/B6) ? A4 Rally
Review | P9 | ? owner | ? | ? |
138
139 **Milestone rollup:** *M0/MVP = P1?P5 ? COMPLETE (2026-06-20)** (pick?reel on existing
API, zero new engine endpoints) . **M1 = P6?P7** (RESTful + reliability polish) . **M2 = P8**
(multi-video home) . **M3 = P9** (per-rally preview/timeline) . **M4 = P10** (corpus correction
loop) . **M5** = swap `parseQuery()` for localized NL behind the same seam (gated on shot/player
data landing).
140
141 ## 8. Open questions ? owner / external
142 **Blocking gates ? ? RESOLVED 2026-06-20 (owner) ? see D8?D11 in §2:**
143 1. ~~Default selection ON vs OFF?~~ ? **ON / curate-by-removal** (D8).
144 2. ~~Is preview-before-select a ship-blocker?~~ ? **No**;; text-metadata MVP, preview is
P9 (D9).
145 3. ~~Job-tethered vs multi-video MVP?~~ ? **Job-tethered** (D10).
146 4. ~~`min_shot_count` override?~~ ? **Warn, don't override** in MVP (D11).
147
148 **Still open (nice-to-knows, non-blocking):**
149 5. May user-facing copy call `parseQuery()` "conversational", given conversational QA is
explicitly deferred (`UI_PROPOSAL.md:66`)? Risk of over-promising NL. *Owner positioning/honesty
call.* *(Lean: avoid the word; show the parsed-pill instead.)*
150 6. One screen for query+grid (recommended) vs a separate ? corpus/correction screen for
A4? *Owner positioning call.*
151
152 ## 9. Definition of done
153 - [ ] From a finished job, a user can list rallies, query/select, and download a reel ?
on **existing** endpoints.
154 - [ ] Only honest fields (ordinal / duration / shot_count+confidence-when-present) are
shown; fabricated `server`/`receiver`/`events` are hidden.
155 - [x] `parseQuery()` covers length/count/order/shot_count, greys out unavailable tokens,
shows a visible parsed-pill, and is unit-tested. *(P3)*
156 - [x] The P3 path is **observable**: structured, correlation-id'd logs for parse / apply
/ skip / empty / unavailable-request / manual-edit; unsupported concepts and zero-result queries
are logged loudly, never dropped silently (§4.4). *(P3)*
157 - [x] Selection hook is unit-tested; the `min_shot_count` floor is surfaced, not hidden.
*(P2 hook . P4 floor warning ? `lib/floor.ts` mirrors `main.py:362-383`, exempts no-shot_count
rallies, blocks Build when all-dropped)*
158 - [x] From a finished job, a user can list rallies, query/select, and download a reel ?
on **existing** endpoints. *(P5 ? `?video=<id>` loads via `POST /api/query`; Build ? `POST
/api/compile` ? `` preview + download over `GET /api/highlights/...`; verified e2e
against a stub engine)*
159 - [x] `npm run build` (and `npm test`) clean; the static UI's `data.rallies` bug is gone
? the SPA **supersedes** the static `app.js` for this flow (reads `play_segments`).
160 - [x] §7 tracker + this DoD updated. *(Doc stays live for the owner-gated M1?M5 engine
milestones; not archived ? only the MVP/M0 slice is complete.)*
161
162 ## 10. References
163 **Engine code anchors `[verified this session]`:** `backend/main.py:121-123` (config),
`:274-293` (jobs), `:307-328` (highlights download), `:331-340` (query), `:342-396` (compile +
`min_shot_count` floor + filepath-not-URL); `backend/infrastructure/database.py:124-135`
(play_segments), `:166-168` (dead correction columns), `:418` (no list-videos), `:427-475`
(query read path + events join); `backend/pipeline/segmenters/gemini.py:461-498, 470-485`

```

```
(honest shot_count, server/receiver=None); `backend/pipeline/segmenters/motion.py:191-198`
(fabricated fields); `backend/pipeline/stitcher/compiler.py:80-105, 303, 311-317` (merge/de-dupe
+ entryptpoint); `backend/api/static/app.js:344` (`data.rallies` bug); `config.json`
(`default_segmenter: "gemini"`).
164    **Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`;
`HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now,
model punted). **This repo:** `web/README.md`, `REQUEST_ACCESS_FORM.md`.
165
166    ---
167    ### Changelog
168    - 2026-06-20 ? Initial `DRAFT`. Consolidated three candidate designs (Timeline / Grid /
RallyPicker) into the hybrid recommendation via a multi-agent design pass; every engine-fact
anchor re-verified against source before writing.
169    - 2026-06-20 ? Owner **approved** the recommendations; locked D8?D11 (default ON .
preview not a ship-blocker . job-tethered MVP . warn-don't-override the floor); status `DRAFT ?
IN PROGRESS`; P0 done.
170    - 2026-06-20 ? **UI: lead with rally LENGTH, park shot-count as experimental** (owner).
Rally cards emphasize duration (`m:ss long`); the shot-count chip is flagged amber
"experimental" with a tooltip ("under development, may be inaccurate ? filter by length
instead"); example chips are duration-led (dropped "at least 8 shots"); a note states "rally
length is fully supported, shot-count is experimental". `parseQuery` still accepts shot-count
queries (not broken) but they're de-featured. Public badge `P5`?`alpha`. A test pins that every
example chip parses to a recognized, fully-available (non-shot-count) query. Suite 75 green.
171    - 2026-06-20 ? **P5 shipped ? MVP / M0 COMPLETE.** `App` now loads a real session from
`?video=<id>` (`POST /api/query`, default ALL-selected D8) with a mode banner (demo / loading /
loaded N / error); **Build** ? `compileReel` (`POST /api/compile`) ? `CompileResult` previews
the stitched reel in `<video>` over `GET /api/highlights/{video_id}/{filename}` (client-built
URL ? engine returns a filepath, not a URL) + a download link; `reelFilename()` slugifies to an
engine-safe bare `.mp4`. Loading/building/done/error states, all traced (`rallies.load_`,
`build.start/done/failed`). Verified e2e against a stub engine: load 5 rallies ? real
`/api/config` floor warning ? Build ? `<video>` reached `readyState=4` (decoded a real mp4
through the proxy) + download link correct; all-below-floor selection keeps Build disabled (P4
guard prevents the 400). `reelFilename` unit-tested; suite **73 green**, typecheck + build
clean. **Next: GCP-GPU end-to-end validation (real video ? rallies ? reels).**
172    - 2026-06-20 ? **P4 shipped**: `SelectionFooter` (live summary + reel-name input +
Build) with an **honest `min_shot_count` warning** ? `lib/floor.ts` mirrors the engine floor
(`main.py:362-383`) EXACTLY: a rally is dropped only if it has a numeric `shot_count` below the
floor; rallies *without* `shot_count` are exempt; **all-below ? Build disabled** (the engine
would 400). Config read from `GET /api/config` on mount (graceful offline). 8 floor unit tests;
amber/red warning states verified live in-browser; suite **70 green**, typecheck + build clean.
173    - 2026-06-20 ? **P3 shipped**: deterministic `parseQuery()` (`web/src/lib/query.ts`)
over length/count/order/shot_count + `runQuery`/`sortRallies`/label helpers and the `QueryBar`
(live parsed-pill, example chips, greyed "not available yet" for shot-type/player/score ? never
faked). Writes the shared selection set via a new `useRallySelection().apply`; `RallyGrid`
ordinals made stable so a re-sort doesn't renumber cards. **First-class observability ($4.4):**
structured, correlation-id'd `query.*` / `selection.*` / `rally.toggle` runlog (load
`empty_result`/`apply_failed`; `unavailable_request` demand signal). **Adversarial multi-agent
review** (grammar/honesty/robustness/reuse) ran pre-merge ? 13 confirmed findings folded in,
incl. 4 majors: an honesty leak (`at least 8 smashes` ? fake `length` ? 8s), a selection-wiping
`top 0`, a unit-blind `first 30 seconds`, and self-contradictory pill labels (`first 10` shown
as "top 10"). Live browser CUJ replayed (filter / top-N / honest-greying / untick / no-op /
wipe-closed). Suite **62 green**, typecheck + build clean.
174
```

5. assistant (2026-06-22T09:46:33.136Z)

Now let me find the staleFindings for my doc. Let me search the audit JSON for the PER_RALLY_SELECTION entries.

6. user (2026-06-22T09:46:36.566Z)

=== staleFindings count: 6

```
[
  {
    "doc": "docs/PER_RALLY_SELECTION.md",
```

```

    "claim": "P6 (engine repo) \u2014 GET /api/health + offline banner; /api/compile returns
download_url \u2014 Status \u2610 (Todo, not built); \u00a74.3 table marks GET /api/health as
'new' and the download_url addition as 'modify' (not yet done).",
    "docLocation": "docs/PER_RALLY_SELECTION.md:133 (P7 tracker row P6), :83-84 (\u00a74.3
table)",
    "verdict": "stale",
    "evidence": "backend/main.py:307 `@app.get(\"/api/health\")` is fully implemented;
backend/main.py compile (~line 769) returns
`{\"status\": \"success\", \"filepath\": out_path, \"download_url\": download_url}` where
download_url = `/api/highlights/{video_id}/{out_name}`. Both P6 items are shipped in the
engine.",
    "suggestedFix": "Flip P6 to \u2611 (or split: health=\u2611, download_url=\u2611) and update
\u00a74.3 status from 'new'/'modify' to 'exists'. Note in the doc that the engine now returns
download_url so the SPA no longer needs to hand-build the URL (though it still can).",
    "severity": "high"
  },
  {
    "doc": "docs/PER_RALLY_SELECTION.md",
    "claim": "Compile returns a server-local filepath, NOT a URL, so the SPA constructs this
download URL itself (repeated as a core caveat in \u00a70 TL;DR, \u00a73 Foundation, \u00a74.2
data flow, D11/risk table, changelog).",
    "docLocation": "docs/PER_RALLY_SELECTION.md:12, :44, :73, :116",
    "verdict": "stale",
    "evidence": "backend/main.py compile endpoint now returns `download_url` alongside
`filepath`: `download_url = f\"/api/highlights/{quote(req.video_id)}/{quote(out_name)}\"; return
{\"status\": \"success\", \"filepath\": out_path, \"download_url\": download_url}`. The 'filepath not
a URL' premise that the whole 'SPA must build the URL' design rests on is no longer the full
picture.",
    "suggestedFix": "Update the caveat to: compile returns BOTH a server-local filepath and a
ready-to-use download_url (/api/highlights/...). The SPA may use download_url directly; the
hand-built-URL fallback is now optional, not required.",
    "severity": "high"
  },
  {
    "doc": "docs/PER_RALLY_SELECTION.md",
    "claim": "P8 \u2014 GET /api/videos + identity scoping \u2014 Status \u26d4 owner, \u2610
(not built); \u00a74.3 marks 'GET /api/videos' as 'new', noting 'only get_video(id) exists,
database.py:418'.",
    "docLocation": "docs/PER_RALLY_SELECTION.md:135 (P8 row), :86 (\u00a74.3)",
    "verdict": "stale",
    "evidence": "backend/main.py:374 `@app.get(\"/api/videos\")` \u2192
`db_instance.list_videos(limit=limit)`; backend/infrastructure/database.py:545 `def
list_videos(...)`\u2014. The endpoint and the DB enumerate both exist (added per PACKAGING_PLAN.md
P1). The doc's 'only get_video(id) exists' claim and the 'new' status are wrong; identity
scoping may still be pending but the listing endpoint shipped.",
    "suggestedFix": "Mark the GET /api/videos endpoint as 'exists' in \u00a74.3; in P8 note that
the videos-list endpoint already shipped and only the identity/owner-scoping layer remains
gated.",
    "severity": "high"
  },
  {
    "doc": "docs/PER_RALLY_SELECTION.md",
    "claim": "P3/P4/P5 tracker rows show PR = '(this PR)' as the merge reference.",
    "docLocation": "docs/PER_RALLY_SELECTION.md:130-132",
    "verdict": "stale",
    "evidence": "git log: P3 merged as #28 (d537d19), P4 as #30 (aff8de8), P5 as #31 (8bf200d).
The '(this PR)' placeholders were never backfilled, so a reader cannot find the actual PRs.",
    "suggestedFix": "Replace '(this PR)' with [#28], [#30], [#31] for P3/P4/P5 respectively.",
    "severity": "medium"
  },
  {
    "doc": "docs/PER_RALLY_SELECTION.md",
    "claim": "Engine line anchors in [verified ...] tags: config at backend/main.py:121-123;
jobs :274-293; highlights download :307-328; query :331-340; compile + floor :342-396;
play_segments database.py:124-135; dead correction cols :166-168; no list-videos :418; query

```

```

read :427-475.",
  "docLocation": "docs/PER_RALLY_SELECTION.md:38-46, :163 (\u00a710 anchors)",
  "verdict": "stale",
  "evidence": "Live engine: /api/config is at main.py:297 (not 121-123); /api/jobs/{job_id} at :637 (not 274-293); /api/highlights at :670 (not 307-328); /api/query at :694 (not 331-340); /api/compile at :705 (not 342-396); play_segments CREATE TABLE at database.py:186 (not 124-135); get_video at :536 (not 418). The endpoints/contracts still EXIST and behave as described, but every cited line number has drifted because the engine ships PRs independently.",
  "suggestedFix": "Re-run the anchors against current engine HEAD and update line numbers, or replace brittle line numbers with symbol names (e.g. main.py::get_config, database.py::list_videos). The sibling DRY_RUN_CUJ.md \u00a78 already warns that engine-surface claims drift between sessions.",
  "severity": "medium"
},
{
  "doc": "docs/PER_RALLY_SELECTION.md",
  "claim": "References \u00a70.6 / \u00a710 cite 'REQUEST_ACCESS_FORM.md' (repo-root style) as a peer doc.",
  "docLocation": "docs/PER_RALLY_SELECTION.md:6, :164",
  "verdict": "stale",
  "evidence": "git ls-files shows the file is at docs/REQUEST_ACCESS_FORM.md, not at repo root. The bare 'REQUEST_ACCESS_FORM.md' reference is path-imprecise (it now lives under docs/).",
  "suggestedFix": "Reference it as docs/REQUEST_ACCESS_FORM.md.",
  "severity": "low"
}
]

```

7. assistant (2026-06-22T09:46:38.931Z)

Now I have my worklist of 6 stale findings. Let me verify each against the live engine repo (origin/master). Let me check the key engine symbols.

8. user (2026-06-22T09:46:43.998Z)

```

=== /api/health ===
origin/master:backend/main.py:307:@app.get("/api/health")
=== /api/capabilities ===
origin/master:backend/main.py:340:@app.get("/api/capabilities")
=== /api/config ===
origin/master:backend/main.py:297:@app.get("/api/config")
=== /api/videos ===
origin/master:backend/main.py:374:@app.get("/api/videos")
=== /api/compile ===
origin/master:backend/main.py:705:@app.post("/api/compile")
=== /api/query ===
origin/master:backend/main.py:694:@app.post("/api/query")
=== /api/jobs ===
origin/master:backend/main.py:611:@app.get("/api/jobs/{job_id}/cost")
origin/master:backend/main.py:637:@app.get("/api/jobs/{job_id}")
=== /api/highlights ===
origin/master:backend/main.py:670:@app.get("/api/highlights/{video_id}/{filename}")

```

9. assistant (2026-06-22T09:46:46.427Z)

The endpoints exist on origin/master. Now let me verify the compile endpoint returns `download\_url`, the `list\_videos` DB method, and the play_segments/get_video locations in database.py.

10. user (2026-06-22T09:46:48.051Z)

```

=== compile endpoint body (download_url) ===
@app.post("/api/compile")
def compile_highlights(req: CompileRequest):

```



```

video = db_instance.get_video(req.video_id)
if not video:
    raise HTTPException(status_code=404, detail="Indexed video record not found.")

# Retrieve matching play segments from db
all_segments = db_instance.query_play_segments(req.video_id, {})
matched_segments = [s for s in all_segments if s["id"] in req.segment_ids]

if not matched_segments:
    raise HTTPException(status_code=400, detail="No matching play segments found to
stitch.")

# Reject path traversal / absolute output names (os.path.join would otherwise let an
# absolute or '..'-laden output_filename write anywhere on disk).
try:
    out_name = safe_output_filename(req.output_filename)
except ValueError as e:
    raise HTTPException(status_code=400, detail=str(e)) from e

# The configured shot-count floor (stitching.min_shot_count) applies on the API
# path too ? parity with scripts/run_process_and_stitch.py. Segments WITHOUT shot_count
# metadata are exempt: the floor filters known counts only, so explicitly
# requested manual/legacy segments can't silently vanish under the default floor.
stitching = getattr(global_config, "stitching", None) or StitchingConfig()
kept = []
for s in matched_segments:
    meta = s.get("metadata") or {}
    sc = meta.get("shot_count") if isinstance(meta, dict) else None
    try:
        if sc is not None and int(sc) < stitching.min_shot_count:
            continue
    except (TypeError, ValueError):
        pass # unparseable count -> treat as unknown, keep
    kept.append(s)
if len(kept) < len(matched_segments):
    logger.info(f"[compile] stitching.min_shot_count={stitching.min_shot_count}: dropped "
        f"{len(matched_segments) - len(kept)}/{len(matched_segments)} segments below
the floor.")
if not kept:
    raise HTTPException(status_code=400,
        detail=f"All {len(matched_segments)} matching segments fall below "
        f"stitching.min_shot_count={stitching.min_shot_count}.")

# Extract start/end time pairs
time_pairs = [(s["start_time"], s["end_time"]) for s in kept]

# Run compiler with the configured stitching paddings + optional branding watermark,
localized to
# the UI locale the SPA rendered in (req.locale): the reel's watermark reads in the user's
language
# with a script-appropriate font. Unknown/absent locale keeps the configured default
unchanged.
output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output") or
"output"
localized_watermark = localize_watermark(stitching.watermark, req.locale)
compiler = VideoCompiler(output_dir, begin_padding=stitching.begin_padding,
    end_padding=stitching.end_padding, watermark=localized_watermark)
try:
    # The stored `filepath` is the SOURCE ref ? a bucket/Drive URI (gs://, s3://, gdrive://)
for a
    # video ingested via /api/process from a URI, or a plain local path. Resolve it to a
LOCAL file
    # the stitcher can read: identity for a local path (byte-identical to before), fetch-to-
scratch
    # for a bucket URI (the same seam the process path uses, _execute_processing

```

```

main.py:159).
    # Without this, compiling a bucket-ingested video 500s ? ffmpeg can't open a gs:// path.
    local_src = storage.acquire_local_input(video["filepath"], getattr(global_config,
"storage", None))
    out_path = compiler.compile_highlights(local_src, time_pairs, out_name)
    # `filepath` is server-local (not reachable by the browser); also return the ready-to-
use
    # download URL so the SPA needn't hand-build it (P6 / PER_RALLY_SELECTION.md §4.3).
    download_url = f"/api/highlights/{quote(req.video_id)}/{quote(out_name)}"
    return {"status": "success", "filepath": out_path, "download_url": download_url}
except Exception as e:
    raise HTTPException(status_code=500, detail=f"Highlight stitching failed: {str(e)}")
from e

```

--- CLI Debug Utility & Orchestration ---

```

def run_cli_diagnose_clip(video_path: str, timestamp: float):
    """CLI debugging utility to examine segment properties around a timestamp."""
    print("=" * 60)
    print(f"DEBUGGING VIDEO CLIP: {video_path} at {timestamp}s")
    print("=" * 60)

    cfg = load_config() # legacy wrapper still works, returns dict for now

    # 1. Validation check diagnostics
    print("[DIAGNOSTIC] Running validation framework...")
    results = registry.run_all(video_path, cfg)
    for r in results:
        status_symbol = "[PASS]" if r.passed else "[FAIL]"
        print(f" {status_symbol} {r.validator_name}: {r.message}")
        if r.details:
            print(f"      Details: {r.details}")

```

11. user (2026-06-22T09:46:49.328Z)

```

=== list_videos + get_video + play_segments in database.py ===
origin/master:backend/infrastructure/database.py:186:      CREATE TABLE IF NOT EXISTS
play_segments (
origin/master:backend/infrastructure/database.py:228:      confirmed_segment_id
INTEGER,
origin/master:backend/infrastructure/database.py:229:      human_label TEXT,
origin/master:backend/infrastructure/database.py:230:      notes TEXT,
origin/master:backend/infrastructure/database.py:233:      FOREIGN KEY
(confirmed_segment_id) REFERENCES play_segments(id) ON DELETE SET NULL
origin/master:backend/infrastructure/database.py:506:      "UPDATE candidate_segments SET
confirmed_segment_id = ? WHERE id = ?",
origin/master:backend/infrastructure/database.py:522:      query += " AND human_label IS
NULL"
origin/master:backend/infrastructure/database.py:536:      def get_video(self, video_id: str) ->
Optional[Dict[str, Any]]:
origin/master:backend/infrastructure/database.py:545:      def list_videos(self, limit:
Optional[int] = None) -> List[Dict[str, Any]]:
origin/master:backend/infrastructure/database.py:774:      def get_video_cost(self, video_id: str)
-> Dict[str, Any]:

```

12. assistant (2026-06-22T09:46:53.285Z)

All three "high" findings are confirmed:

1. `/api/health` exists (main.py:307) and compile returns `download\_url`.
2. compile returns BOTH `filepath` and `download\_url`.
3. `/api/videos` exists (main.py:374) and `list\_videos` exists (database.py:545).

Let me verify the remaining structural anchors cited in the doc (config, query, jobs, highlights line numbers) and the git PR references for P3/P4/P5, plus the REQUEST_ACCESS_FORM.md location.

13. user (2026-06-22T09:46:56.016Z)

```
=== query_play_segments / play_segments columns context ===
CREATE TABLE IF NOT EXISTS play_segments (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    video_id TEXT NOT NULL,
    start_time REAL NOT NULL,
    end_time REAL NOT NULL,
    duration REAL NOT NULL,
    server TEXT,
    receiver TEXT,
    metadata TEXT,
    FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE
)
"""

# Events Table
cursor.execute("""
CREATE TABLE IF NOT EXISTS events (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    segment_id INTEGER NOT NULL,
    timestamp REAL NOT NULL,
    type TEXT NOT NULL,
    player TEXT,
    details TEXT,
    FOREIGN KEY (segment_id) REFERENCES play_segments(id) ON DELETE CASCADE
)
""")

=== get_video line ===
origin/master:backend/infrastructure/database.py:536:     def get_video(self, video_id: str) ->
Optional[Dict[str, Any]]:
=== query read path / events join ===
origin/master:backend/infrastructure/database.py:564:     def query_play_segments(self, video_id:
str, filters: Dict[str, Any]) -> List[Dict[str, Any]]:
```

14. user (2026-06-22T09:46:58.389Z)

```
=== P3/P4/P5 PR git log ===
05915f2 docs(next-steps): per-rally MVP/M0 COMPLETE (P1?P5, #28/#30/#31); CUJ screencast + M1?M5
next (#32)
8bf200d feat(web): per-rally MVP P5 ? load real rallies + Build?compile?preview?download (MVP
complete) (#31)
aff8de8 feat(web): per-rally MVP P4 ? SelectionFooter + honest min_shot_count warning (#30)
66805ab docs(next-steps): per-rally P3 (parseQuery query bar) merged (#28); P4 next (#29)
d537d19 feat(web): per-rally MVP P3 ? deterministic parseQuery() query bar (traced +
adversarially hardened) (#28)
=== REQUEST_ACCESS_FORM location ===
docs/REQUEST_ACCESS_FORM.md
```

15. assistant (2026-06-22T09:47:04.438Z)

All findings confirmed against the live repos:

- **P3 = #28** (d537d19), **P4 = #30** (aff8de8), **P5 = #31** (8bf200d) ? confirmed in khelsutra git log.
- `REQUEST\_ACCESS\_FORM.md` is at `docs/REQUEST\_ACCESS\_FORM.md`.
- Engine line numbers have all drifted (config :297, jobs :637, highlights :670, query :694, compile :705; database.py play_segments :186, dead cols :228-230, get_video :536, query read :564).

Note: the doc's `play\_segments` column list says `id, video\_id, start\_time, end\_time, duration, server, receiver, metadata` ? matches the live schema exactly (the doc just had wrong line numbers). Also the doc claims dead correction columns are in `candidate\_segments` at database.py:166-168 ? on live master they're in the play_segments CREATE TABLE block at :228-230. Let me confirm where those columns actually live.

16. user (2026-06-22T09:47:05.523Z)

```
=== candidate_segments table + its columns ===
# so a new segmenter reuses this table by setting its own `detector`/`stats`.
cursor.execute("""
    CREATE TABLE IF NOT EXISTS candidate_segments (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        video_id TEXT NOT NULL,
        detector TEXT NOT NULL,
        chunk_index INTEGER,
        start_time REAL NOT NULL,
        end_time REAL NOT NULL,
        duration REAL NOT NULL,
        confidence REAL,
        stats TEXT,
        detection_params TEXT,
        confirmed_segment_id INTEGER,
        human_label TEXT,
        notes TEXT,
        created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
        FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE,
        FOREIGN KEY (confirmed_segment_id) REFERENCES play_segments(id) ON DELETE SET
NULL
    )
""")
```

17. assistant (2026-06-22T09:47:10.675Z)

Confirmed: `candidate\_segments.confirmed\_segment\_id/human\_label/notes` are the dead correction columns, at database.py:227-230. The doc's claim that these columns exist on `candidate\_segments` is CORRECT (only the line number :166-168 drifted).

Now I have everything verified. Let me apply the surgical edits, starting with the symbol-anchor approach for the engine line numbers (per the instructions, prefer symbol-based references for brittle anchors). Let me begin with the highest-severity findings.

****Finding 1 & 2: compile returns `download\_url` now.**** Update §0 TL;DR, §3 Foundation, §4.2, §4.3 table, P6 row, D11/risk table.

Let me start with the §3 Foundation download bullet.

18. user (2026-06-22T09:47:17.981Z)

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19. assistant (2026-06-22T09:47:20.421Z)

Let me verify the actual function names in main.py so my symbol references are accurate.

20. user (2026-06-22T09:47:21.855Z)

```
297:@app.get("/api/config")
298:def get_config():
307:@app.get("/api/health")
308:def health():
--
374:@app.get("/api/videos")
375:def list_videos(limit: Optional[int] = None):
--
611:@app.get("/api/jobs/{job_id}/cost")
612:def get_job_cost(job_id: str):
626:@app.get("/api/videos/{video_id}/cost")
627:def get_video_cost(video_id: str):
```

```

637:@app.get("/api/jobs/{job_id}")
638:def get_job(job_id: str):
670:@app.get("/api/highlights/{video_id}/{filename}")
671:def download_highlight(video_id: str, filename: str):
694:@app.post("/api/query")
695:def query_play_segments(req: QueryRequest):
705:@app.post("/api/compile")
706:def compile_highlights(req: CompileRequest):

```

21. assistant (2026-06-22T09:47:24.957Z)

My symbol names are correct: `get\_config`, `health`, `list\_videos`, `get\_job`, `download\_highlight`, `query\_play\_segments`, `compile\_highlights`. Let me clean up the §3 edit ? the nested-backtick parentheticals are awkward markdown. Let me simplify to clean symbol references.

22. user (2026-06-22T09:47:31.241Z)

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23. assistant (2026-06-22T09:47:35.159Z)

Now the §0 TL;DR. It references the endpoint anchors and the "filepath not a URL" caveat indirectly. Let me update the engine-anchor line numbers there and the caveat. Let me look at the exact text again.

The §0 TL;DR line: `POST /api/query ? POST /api/compile ? GET /api/highlights [verified backend/main.py:331-340, 342-396, 307-328]`. Let me update those to symbol refs.

24. user (2026-06-22T09:47:37.829Z)

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25. assistant (2026-06-22T09:47:41.219Z)

Now §1 Problem & goal has anchors `[verified backend/main.py:362-383]`, `[verified backend/api/static/app.js:344 vs backend/main.py:340]`. Let me verify the static app.js bug location and the min_shot_count floor still applies in compile_highlights (it does, I saw it). The floor logic is now inline in `compile\_highlights`. Let me update these.

26. user (2026-06-22T09:47:44.969Z)

```

=== static app.js data.rallies bug ===
origin/master:backend/api/static/app.js:344:            renderRallyCards(data.rallies || []);
=== gemini.py shot_count / server None ===
origin/master:backend/pipeline/segmenters/gemini.py:369:    def _validate_and_dedupe(self,
ai_segments: List[Dict[str, Any]], video_duration: float) -> List[Dict[str, Any]]:
origin/master:backend/pipeline/segmenters/gemini.py:480:        shot_count =
segment.get("shuttle_exchange_count")
origin/master:backend/pipeline/segmenters/gemini.py:492:        "shot_count_confidence":
segment.get("shot_count_confidence", "Unknown"),
=== motion.py fabricated fields ===
origin/master:backend/pipeline/segmenters/motion.py:3:import random
origin/master:backend/pipeline/segmenters/motion.py:181:    For each detected segment this
assigns a random server/receiver, fabricates
origin/master:backend/pipeline/segmenters/motion.py:183:    with the stdlib ``random``
module (the legacy demo's fabricated baseline).
origin/master:backend/pipeline/segmenters/motion.py:184:    Returns the per-segment dicts
handed back to the caller. The ``random``
origin/master:backend/pipeline/segmenters/motion.py:190:        # Assign Server and Receiver
randomly from pool
origin/master:backend/pipeline/segmenters/motion.py:191:        server, receiver =

```

```

random.sample(player_pool, 2)
origin/master:backend/pipeline/segmenters/motion.py:196:         "shot_count":
random.randint(4, 25),
origin/master:backend/pipeline/segmenters/motion.py:198:         "avg_speed_kmh":
round(random.uniform(40, 85), 1)
origin/master:backend/pipeline/segmenters/motion.py:206:         serve_foul =
random.random() < 0.15 # 15% chance of serve foul
origin/master:backend/pipeline/segmenters/motion.py:212:         details={"status":
"foul" if serve_foul else "valid", "type": "low_serve" if random.random() > 0.5 else
"high_serve"}
=== compiler.py merge ===
origin/master:backend/pipeline/stitcher/compiler.py:83:     def _merge_overlapping(segments:
List[Tuple[float, float]]) -> List[Tuple[float, float]]:
origin/master:backend/pipeline/stitcher/compiler.py:305:     def compile_highlights(self,
raw_video_path: str, segments: List[Tuple[float, float]], output_filename: str) -> str:
origin/master:backend/pipeline/stitcher/compiler.py:316:         segments =
self._merge_overlapping(segments)

```

27. assistant (2026-06-22T09:47:51.283Z)

The audit's worklist only flagged 6 specific findings. The line-number drift finding (finding 5) cites a specific set: `config :121-123; jobs :274-293; highlights :307-328; query :331-340; compile :342-396; play\_segments database.py:124-135; dead correction cols :166-168; no list-videos :418; query read :427-475` and §10 anchors. I should update those specific anchors but stay within the finding's scope. The finding's suggestedFix says to update line numbers OR replace with symbol names. I'll use symbol names where there's a clean symbol, and keep it surgical.

Let me handle finding 5 systematically. First, §3 Foundation remaining anchors (the rally store, listing rallies, building a reel bullets).

28. user (2026-06-22T09:47:52.679Z)

```

41     - **The rally store is `play_segments`** ? columns `id, video_id, start_time, end_time,
duration, server, receiver, metadata` `[verified database.py:124-135]`. `id` is the integer the
UI selects and passes to compile; `start_time/end_time/duration` are REAL seconds (duration
computed at insert).
42     - **Listing rallies = `POST /api/query {video_id}` ? `{play_segments:[...]}`** ? the
*only* rally-list path today; it JSON-parses `metadata` and joins `events` `[verified
main.py:331-340, database.py:427-475]`.
43     - **Building a reel = `POST /api/compile {video_id, segment_ids:[int],
output_filename}`** ? loads `videos.filepath`, filters all segments to the requested
`segment_ids`, applies the `min_shot_count` floor, extracts `time_pairs=[(start,end)]`, and
calls `VideoCompiler.compile_highlights(raw_path, segments, name)` `[verified main.py:342-396,
compiler.py:303]`, which **merges overlaps** then per-clip re-encodes ? concat ? one MP4
`[verified compiler.py:311-317]`. Selection order/duplicates are therefore safe.
44     - **Downloading = `GET /api/highlights/{video_id}/{filename}`** ? streams the compiled
MP4. Compile now returns **both** a server-local `filepath` **and** a ready-to-use
`download_url` (`/api/highlights/...`) `[verified main.py::compile_highlights]`, so the SPA may
use `download_url` directly; the hand-built-URL fallback is now optional, not required. The
compiled reel is the **only streamable video that exists today** `[verified
main.py::download_highlight]`.
45     - **The floor knob is readable** ? `GET /api/config` exposes `stitching.min_shot_count`
`[verified main.py::get_config]`, so the UI can warn before a selected rally is silently
dropped.
46     - **Job entry** ? `GET /api/jobs/{job_id}` returns `result.video_id` `[verified
main.py::get_job]`, which is how the MVP reaches a video; a list-videos endpoint (`GET
/api/videos`) has since shipped (§4.3).
47
48     - **Honest fields vs fabricated/roadmap (critical for the UI):**
49     - **Honest today** (default `default_segmenter: "gemini"`, per `config.json`
`[verified]`): `start_time`, `end_time`, `duration`, and `metadata.shot_count` (Gemini's
`shuttle_exchange_count`, fallback `max(3, int(dur/0.8))`) + `metadata.shot_count_confidence` (a
**string**, often "Unknown") `[verified gemini.py:470-485]`. On the gemini path
`server`/`receiver` are written `None` ("no fabricated data") `[verified gemini.py:480-487]`.

```

50 - ****Fabricated ? do NOT surface:**** the `motion` demo segmenter fills `server`/`receiver` via `random.sample(...)` , a random `shot\_count` , and random `events` (serve/smash/foul/winner) `[verified motion.py:191-198]` . `query` returns these columns + `events` , but they are not ground truth and must stay hidden.

51 - ****Roadmap ? no column, no detector:**** shot-type per shot, player identity, match/point score, game/set grouping. None exist anywhere in the schema; do not consume them `[verified database.py whole-schema 122-148]` .

52 - ****Correction-loop columns exist but are dead:**** `candidate\_segments.human\_label/notes/confirmed\_segment\_id` are defined but no endpoint writes them `[verified database.py:166-168]` ? that's the M4 corpus extension (engine PR-4/B6, `UI\_ALPHA\_EXECUTION\_PLAN.md`) .

29. assistant (2026-06-22T09:47:57.970Z)

The play_segments column list in the doc matches the live schema. The query read path is `query\_play\_segments` in database.py (the DB method). Let me update these anchors to symbol references. Note `compiler.py:303` ? `compile\_highlights` is at :305; `compiler.py:311-317` ? the merge is at :316. I'll use symbol refs for these.

30. user (2026-06-22T09:48:06.391Z)

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31. assistant (2026-06-22T09:48:09.881Z)

Now the dead correction columns anchor (`database.py:166-168`) ? these live in the `candidate\_segments` CREATE TABLE. The doc text already correctly says `candidate\_segments.human\_label/notes/confirmed\_segment\_id` . Let me update the anchor.

32. user (2026-06-22T09:48:12.764Z)

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33. assistant (2026-06-22T09:48:23.534Z)

The gemini.py anchors (`:470-485`, `:480-487`) and motion.py (`:191-198`) and database.py whole-schema (`122-148`) ? the audit finding 5 listed a specific set of anchors to fix and did NOT include gemini.py/motion.py/static app.js in its enumerated list. The finding's enumerated anchors are: config, jobs, highlights, query, compile, play_segments, dead correction cols, no list-videos, query read. To stay surgical and within the finding's scope, I'll leave gemini.py/motion.py anchors as-is (they weren't flagged, and verifying they drifted isn't part of this finding).

Actually, let me re-read finding 5's scope: it lists `docLocation: docs/PER\_RALLY\_SELECTION.md:38-46, :163 (\$10 anchors)` . That's \$3 (lines 38-46) and \$10. The gemini/motion lines (49-50) are outside :38-46. So I'll keep my edits within \$3 lines 41-46 and \$10. Good ? I've handled \$3.

Now the "no list-videos `:418`" anchor ? this is finding 3's territory (the `GET /api/videos` now exists). It appears in \$4.3 table line 86. Let me handle finding 3 (P8 / \$4.3 GET /api/videos) and finding 1 (\$4.3 health + download_url, P6 row) together. Let me re-read the \$4.3 table.

34. user (2026-06-22T09:48:24.337Z)

75 ****4.3 Engine API additions (none for MVP; tagged against real gaps)****

76 | Endpoint | Purpose | Status |

77 |---|---|---|

78 | `POST /api/query` | List rallies (only path) ? fix SPA to read `play\_segments` |

****exists**** |

79 | `POST /api/compile` | Stitch selected `segment\_ids` ? MP4 | ****exists**** |

80 | `GET /api/highlights/{video\_id}/{filename}` | Stream/download compiled reel |

```

**exists** |
81      | `GET /api/config` | Read `stitching.min_shot_count` for the warning | **exists** |
82      | `GET /api/jobs/{job_id}` | Carry `result.video_id` into the picker | **exists** |
83      | `POST /api/compile` ? add `download_url` | Return `/api/highlights/...` so SPA stops
hand-building it (additive, low-risk; today returns only a filepath `main.py:394`) | **modify**
|
84      | `GET /api/health` | Engine-offline banner ping the SPA is specified to use
(`HOSTING_PLAN.md:42`) | **new** |
85      | `GET /api/videos/{video_id}/rallies` | RESTful/cacheable alias returning the same
`{play_segments}` shape | **new** |
86      | `GET /api/videos` | List a user's videos for a Match picker (only `get_video(id)`
exists, `database.py:418`) ? needs DB enumerate + identity scoping | **new** |
87      | `GET /api/videos/{video_id}/source` (Range) *or* `/clip?start=&end=` | Stream
source/per-span video so a rally previews before selection ? neither exists | **new** |
88      | `GET /api/videos/{video_id}/rallies/{segment_id}/thumbnail` | Per-rally poster still
(no thumbnail column; WASB frames are transient) | **new** |
89      | `PATCH /api/segments/{id}` | Persist accept/reject/nudge ?
`candidate_segments.human_label/notes/...` (columns exist, unwritten ? corpus loop) | **new** |
90

```

35. assistant (2026-06-22T09:48:27.727Z)

Now I'll update the §4.3 table rows: the `download\_url` modify?exists, `GET /api/health` new?exists, and `GET /api/videos` new?exists (listing shipped, only scoping pending). Let me verify `GET /api/videos/{video\_id}/rallies` does NOT exist (so it stays "new").

36. user (2026-06-22T09:48:30.067Z)

```

=== /api/videos/{video_id}/rallies alias ===
=== PATCH /api/segments ===
=== /source or /clip ===

```

37. assistant (2026-06-22T09:48:33.492Z)

Confirmed: `GET /api/videos/{video\_id}/rallies`, `PATCH /api/segments/{id}`, source/clip/thumbnail all still absent ? those rows stay "new". Now apply the §4.3 table edits.

38. user (2026-06-22T09:48:39.251Z)

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39. assistant (2026-06-22T09:48:44.394Z)

Now finding 1 & 2 in §4.2 data flow and §0 ? the "filepath not a URL" caveat. Let me look at §4.2 line 73 and §0 line 12.

§0 line 12 final sentence: "the *final compiled reel* can stream" ? that part doesn't mention filepath. The "filepath not a URL" in §0 was via the endpoint anchors I already updated. Let me check line 73 (§4.2) and line 116 (risk table D11 reference).

40. user (2026-06-22T09:48:45.168Z)

```

66      **4.2 Data flow** `[verified anchors inline]`
67      Gemini segmenter (process time) writes `play_segments {id, start/end_time, duration,
metadata.shot_count/confidence/detector}` `[gemini.py:461-498]`
68      ? SPA `POST /api/query {video_id}` ? `{play_segments:[...]}` `[main.py:331-340]`
(**fix** the static UI's `data.rallies` read ? `play_segments` `[app.js:344]`)
69      ? normalize to `RallyCard[]` + one shared `selectedIds:Set`;
`server`/`receiver`/`events` are present in the payload but motion-only fabrications ? **not
surfaced** `[motion.py:191-198]`
70      ? all input modes mutate the **same** `selectedIds`: (a) `parseQuery()` predicate over
duration/ordinal/shot_count; (b) checkbox toggle + bulk ops; (c) *(post-MVP)* timeline band
71      ? footer recomputes count + ?duration **client-side**, and warns for any selected rally

```



```

with `shot_count < stitching.min_shot_count` `[main.py:362-383]`
72      ? **Build** ? `POST /api/compile {video_id, segment_ids: [...selectedIds],
output_filename}` ? stitcher de-dupes via `_merge_overlapping` `[compiler.py:311-317]`, cuts
`time_pairs` from `videos.filepath`, re-encodes per clip ? concat ? one MP4 `[main.py:386-393,
compiler.py:303]`
73      ? compile returns a **server-local filepath, not a URL** `[main.py:394]` ? SPA builds
`GET /api/highlights/{video_id}/{output_filename}` to preview in `` + download
`[main.py:307-328]`.

```

41. assistant (2026-06-22T09:48:52.736Z)

The §4.2 anchors (lines 67-73) are within finding 2's docLocation set partially, but finding 5 lists `:38-46, :163` only. Finding 2's docLocation is `:12, :44, :73, :116`. Line 73 IS in finding 2's scope. Line 73's "compile returns a server-local filepath, not a URL ? SPA builds..." is the exact caveat finding 2 targets. Let me update line 73.

The §4.2 line 73 has anchors `[main.py:394]` and `[main.py:307-328]`. I'll update the caveat substance and the anchor.

42. user (2026-06-22T09:48:55.269Z)

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43. assistant (2026-06-22T09:48:57.879Z)

Now finding 2's line 116 ? the risk table row "compile returns a filepath not a URL". Let me read the risk table and D11.

44. user (2026-06-22T09:48:58.522Z)

```

108      ## 5. Risk table
109      | Risk | Severity | Mitigation |
110      |---|---|---|
111      | **No source/clip/frame/thumbnail endpoint** ? only compiled reels stream
`[main.py:307-328]` | High (UX) | MVP selects on text metadata + previews the *final* reel;
preview/timeline deferred to M3 behind a new endpoint |
112      | `min_shot_count` silently drops selected rallies `[main.py:362-383]` | High (looks
broken) | Footer warns *before* Build; D7 makes selection explicit; an override is an owner gate
(Launch-Report-worthy `[memory: launch-report-convention]`) |
113      | No match-duration field ? a timeline must derive `total = max(end_time)`, mis-
rendering pre/post dead air | Med | Timeline deferred to M3; never over-promise it as a true
scrubber until source streams |
114      | `shot_count` is an *estimate* (`shuttle_exchange_count` w/ `max(3,int(dur/0.8))`
fallback) `[gemini.py:470-485]` | Med | Label "approx."; confidence shown as the raw string, not
a meter |
115      | No auth / user scoping anywhere (CORS `*`, `videos.user_id` never set/filtered) | High
for multi-tester | Fine for single-box local alpha; a ship-blocker for any shared host ? gate
`GET /api/videos` behind PR-3 scoping |
116      | compile returns a filepath not a URL `[main.py:394]` | Low | Client builds the URL for
MVP; `download_url` field (M1) removes the brittleness |
117      | Surface sprawl (query + grid + future timeline + future correction) | Med | Deliberate
IA: query+grid = one screen; correction = separate ? corpus screen (owner call, §8) |

```

45. assistant (2026-06-22T09:49:02.861Z)

Line 116 is the risk-table row finding 2 targets. Now the `download\_url` field has shipped, so the risk is resolved. Let me update it. (The `[main.py:307-328]` and `[main.py:362-383]` anchors on lines 111-112 are within the risk table but NOT in finding 5's enumerated `:38-46, :163` scope, nor finding 2's `:12,:44,:73,:116`. I'll leave 111-112 untouched to avoid scope creep ? they remain accurate descriptions even if line-drifted, and weren't flagged.)

Let me update line 116.

46. user (2026-06-22T09:49:05.452Z)

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47. assistant (2026-06-22T09:49:08.883Z)

Now finding 4 + finding 1 (P6 row) ? the \$7 tracker. Let me update P3/P4/P5 "(this PR)" ? #28/#30/#31, and P6 ????. Let me re-read the tracker rows.

48. user (2026-06-22T09:49:09.821Z)

```
125      | ID | Deliverable | Depends on | Gated? | Status | PR |
126      |----|-----|-----|-----|-----|----|
127      | P0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | No | ? |
128      | P1 | SPA scaffold (Vite+React+TS+Tailwind) in `web/` + typed API client; dev proxy to
engine `:8000` | P0 | No | ? | [#14](https://github.com/avidullu/khelsutra/pull/14) |
129      | P2 | RallyList/Grid (honest fields only) + `useRallySelection()` hook (unit-tested) |
P1 | No | ? | [#21](https://github.com/avidullu/khelsutra/pull/21) |
130      | P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable
tokens) | P2 | No | ? | (this PR) |
131      | P4 | SelectionFooter + `min_shot_count` warning (from `GET /api/config`) | P2 | No | ?
| (this PR) |
132      | P5 | Build ? `POST /api/compile` ? client-built download URL + `` preview of
the reel | P2 | No | ? | (this PR) |
133      | P6 | *(engine repo)* `GET /api/health` + offline banner; `/api/compile` returns
`download_url` | ? | No | ? | ? |
134      | P7 | *(engine repo)* `GET /api/videos/{id}/rallies` RESTful alias | ? | No | ? | ? |
135      | P8 | *(engine repo)* `GET /api/videos` + PR-3 identity scoping ? A1 Matches screen |
P7 | ? owner | ? | ? |
136      | P9 | *(engine repo)* source/clip stream + timeline + per-rally preview/thumbnail | P8
| ? owner | ? | ? |
137      | P10 | *(engine repo)* `PATCH /api/segments/{id}` corpus loop (PR-4/B6) ? A4 Rally
Review | P9 | ? owner | ? | ? |
138
139      **Milestone rollout:** **M0/MVP = P1?P5 ? COMPLETE (2026-06-20)** (pick?reel on existing
API, zero new engine endpoints) . **M1 = P6?P7** (RESTful + reliability polish) . **M2 = P8**
(multi-video home) . **M3 = P9** (per-rally preview/timeline) . **M4 = P10** (corpus correction
loop) . **M5** = swap `parseQuery()` for localized NL behind the same seam (gated on shot/player
data landing).
```

49. assistant (2026-06-22T09:49:13.625Z)

Now update P3/P4/P5 PR cells, and P6 status. For P6: BOTH items (`GET /api/health` AND `/api/compile` returns `download\_url`) are shipped on the engine. The offline banner is the SPA-side consumer ? let me check whether the SPA consumes `/api/health` for an offline banner, since P6 also includes "+ offline banner".

50. user (2026-06-22T09:49:15.785Z)

```
=== SPA health usage ===
web/src/api/client.test.ts:109:    vi.stubGlobal("fetch", vi.fn(async () => { throw new
Error("offline"); } }) as unknown as FetchFn);
web/src/components/IngestPanel.test.tsx:85:        throw new Error("offline");
web/src/components/QueryBar.test.tsx:4:// live parsed-pill, and the "unreadable" branch were
untested (funcs 40%). The offline parser
web/src/components/SetupWizard.test.tsx:46:    it("checks capabilities, then step 1 guides a
missing Gemini key (offline-first default)", async () => {
web/src/components/SetupWizard.test.tsx:98:    it("degrades gracefully when the engine is
unreachable (offline) ? message + a Done escape", async () => {
web/src/components/SetupWizard.test.tsx:103:        // No step content / step indicator when
offline.
web/src/components/SetupWizard.tsx:4:// no key-write endpoint yet, so we tell the user how to
```

```
add the key (offline `motion` works with none).
web/src/components/SetupWizard.tsx:84:  const [offline, setOffline] = useState(false);
web/src/components/SetupWizard.tsx:116:  const stepping = !loading && !offline; // the 3-step
flow only when capabilities loaded
web/src/components/SetupWizard.tsx:142:      {!loading && offline && <p className="text-
amber-700">{t("wizard.offline")}</p>}
=== SPA download_url usage ===
```

51. assistant (2026-06-22T09:49:22.527Z)

The SPA does NOT yet consume `/api/health` for an offline banner, nor `download\_url`. P6 is a multi-part deliverable that mixes engine work (both shipped) and SPA work (offline banner ? not yet wired; the SPA still hand-builds the download URL).

Per the audit finding 1's suggestedFix: "Flip P6 to ? (or split: health=?, download_url=?) and update \$4.3 status... Note in the doc that the engine now returns download_url so the SPA no longer needs to hand-build the URL (though it still can)."

But honesty gate: P6 as written = "`GET /api/health` + offline banner; `/api/compile` returns `download\_url`". The **engine halves are done** but the **SPA offline banner is NOT wired** and the SPA does NOT yet consume `download\_url`. Flipping the whole row to ? would be dishonest. I'll mark it ? (In progress) with the honest split: engine endpoints shipped, SPA consumption pending. This matches the audit's "(or split)" option and keeps the marker honest. Let me apply.

52. user (2026-06-22T09:49:28.645Z)

The file C:\Users\avidu\Projects\khelsutra\docs\PER_RALLY_SELECTION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

53. assistant (2026-06-22T09:49:32.148Z)

Now finding 3's P8 row ? the `GET /api/videos` listing endpoint shipped; only identity scoping remains gated. The P8 row says "(engine repo) `GET /api/videos` + PR-3 identity scoping ? A1 Matches screen". Let me update the row to note the listing endpoint shipped.

54. user (2026-06-22T09:49:34.507Z)

The file C:\Users\avidu\Projects\khelsutra\docs\PER_RALLY_SELECTION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

55. assistant (2026-06-22T09:49:38.150Z)

Now finding 6 ? the `REQUEST\_ACCESS\_FORM.md` references. They appear at \$0.6 (line 6) and \$10 (line 164). Let me check both.

56. user (2026-06-22T09:49:39.348Z)

```
6:> **Relation:** second major tracked work item in the `khelsutra` repo (peer of
`REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
164:**Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`;
`HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now,
model punted). **This repo:** `web/README.md`, `REQUEST_ACCESS_FORM.md`.
```

57. assistant (2026-06-22T09:49:41.736Z)

Both references say bare `REQUEST\_ACCESS\_FORM.md` ? should be `docs/REQUEST\_ACCESS\_FORM.md`. Both are unique enough strings. Let me update both.

58. user (2026-06-22T09:49:43.559Z)

The file C:\Users\avidu\Projects\khelsutra\docs\PER_RALLY_SELECTION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

59. user (2026-06-22T09:49:47.307Z)

The file C:\Users\avidu\Projects\khelsutra\docs\PER_RALLY_SELECTION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

60. assistant (2026-06-22T09:49:51.239Z)

Now finding 5's §10 anchors (line 163). Let me read and update the engine-anchor list in §10 to match the symbol/line-drift fix.

61. user (2026-06-22T09:49:55.316Z)

```
162     ## 10. References
163     **Engine code anchors `[verified this session]`:** `backend/main.py:121-123` (config),
`:`274-293` (jobs), `:`307-328` (highlights download), `:`331-340` (query), `:`342-396` (compile +
`min_shot_count` floor + filepath-not-URL); `backend/infrastructure/database.py:124-135`
(play_segments), `:`166-168` (dead correction columns), `:`418` (no list-videos), `:`427-475`
(query read path + events join); `backend/pipeline/segmenters/gemini.py:461-498, 470-485`
(honest shot_count, server/receiver=None); `backend/pipeline/segmenters/motion.py:191-198`
(fabricated fields); `backend/pipeline/stitcher/compiler.py:80-105, 303, 311-317` (merge/de-dupe
+ entrypoint); `backend/api/static/app.js:344` (`data.rallies` bug); `config.json`
(`default_segmenter: "gemini"`).
164     **Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`;
`HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now,
model punted). **This repo:** `web/README.md`, `docs/REQUEST_ACCESS_FORM.md`.
```

62. assistant (2026-06-22T09:50:02.385Z)

This §10 anchor line is exactly what finding 5 targets (it explicitly cites `:`163` (§10 anchors)). I'll convert the brittle line numbers to symbol references for the main.py/database.py/compiler.py anchors. The finding's enumerated drift covers: config, jobs, highlights, query, compile, play_segments, dead correction cols, no list-videos (now exists), query read.

Note: the gemini.py and motion.py anchors are also here. For gemini.py, the segment-building code is around :480-492; motion.py :191-198 is still accurate (I verified random.sample at :191, shot_count at :196). I'll convert main/database/compiler to symbols and leave gemini/motion/app.js/static as-is or lightly corrected. Let me also note that `:`418` (no list-videos)` is now wrong ? list_videos exists at database.py:545. I'll fix that note.

The static `app.js:344` is still accurate (verified). Let me build the replacement.

63. user (2026-06-22T09:50:09.537Z)

The file C:\Users\avidu\Projects\khelsutra\docs\PER_RALLY_SELECTION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

64. assistant (2026-06-22T09:50:16.684Z)

Now let me address the §3 line 45-46 (already done), and there's one more anchor in §3 line 45 that mentioned `[verified main.py:121-123]` for config ? already updated to `::get\_config`.

Let me also handle the D4/D5/D7 decision-table anchors and the §4 reuse map anchors that have line numbers, since finding 5 says "§10 anchors" but the primary docLocation is `:`38-46` (§3, done) and `:`163`. The decision table (D4 `main.py:342-396`, D5 `database.py:122-135`, D7 `main.py:362-383`) are at lines 29-32, outside the finding's `:`38-46`/`:`163` scope. To stay surgical and avoid scope creep, I'll leave those ? they weren't enumerated in the finding's docLocation.

Let me now also double-check the changelog entries ? finding 2 mentions "changelog" generally, but its docLocation is `:`12, :44, :73, :116` (none in changelog). The changelog is historical

record and should NOT be rewritten (it documents what was true at the time). I'll leave the changelog.

Let me do a final review of all my edits to ensure consistency and no broken markdown.

65. user (2026-06-22T09:50:19.383Z)

```
docs/PER_RALLY_SELECTION.md | 42 ++++++-----
1 file changed, 21 insertions(+), 21 deletions(-)
=== remaining stale anchors check ===
6:> **Relation:** second major tracked work item in the `khelsutra` repo (peer of
`docs/REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
29:| D4 | Compile via existing `POST /api/compile {video_id, segment_ids, output_filename}` |
`[verified backend/main.py:342-396]` | Selection order is harmless ? stitcher
`_merge_overlapping` de-dupes `[verified compiler.py:80-105, 311-317]` |
30:| D5 | Query grammar (alpha) = length, count, order, **shot_count only** | `[verified
database.py:122-135, gemini.py:470-485]` | Only fields the payload actually returns; grey out
the rest |
32:| D7 | Replace the silent `min_shot_count` cut with **explicit user selection** | Owner
framing + `[verified main.py:362-383]` | UI *warns* about the floor; it never hides why a rally
vanished |
71:? footer recomputes count + ?duration **client-side**, and warns for any selected rally with
`shot_count < stitching.min_shot_count` `[main.py:362-383]`
72:? **Build** ? `POST /api/compile {video_id, segment_ids: [...selectedIds], output_filename}`
? stitcher de-dupes via `_merge_overlapping` `[compiler.py:311-317]`, cuts `time_pairs` from
`videos.filepath`, re-encodes per clip ? concat ? one MP4 `[main.py:386-393, compiler.py:303]`
111:| **No source/clip/frame/thumbnail endpoint** ? only compiled reels stream
`[main.py:307-328]` | High (UX) | MVP selects on text metadata + previews the *final* reel;
preview/timeline deferred to M3 behind a new endpoint |
112:| `min_shot_count` silently drops selected rallies `[main.py:362-383]` | High (looks broken)
| Footer warns *before* Build; D7 makes selection explicit; an override is an owner gate
(Launch-Report-worthy `[memory: launch-report-convention]`) |
157:- [x] Selection hook is unit-tested; the `min_shot_count` floor is surfaced, not hidden.
*(P2 hook · P4 floor warning ? `lib/floor.ts` mirrors `main.py:362-383`, exempts no-shot_count
rallies, blocks Build when all-dropped)*
164:**Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`;
`HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now,
model punted). **This repo:** `web/README.md`, `docs/REQUEST_ACCESS_FORM.md`.
172:- 2026-06-20 ? **P4 shipped**: `SelectionFooter` (live summary + reel-name input + Build)
with an **honest `min_shot_count` warning** ? `lib/floor.ts` mirrors the engine floor
(`main.py:362-383`) EXACTLY: a rally is dropped only if it has a numeric `shot_count` below the
floor; rallies *without* `shot_count` are exempt; **all-below ? Build disabled** (the engine
would 400). Config read from `GET /api/config` on mount (graceful offline). 8 floor unit tests;
amber/red warning states verified live in-browser; suite **70 green**, typecheck + build clean.
```

66. assistant (2026-06-22T09:50:25.662Z)

The remaining `main.py:` / `database.py:` line-number anchors (lines 29, 30, 32, 71, 72, 111, 112, 157, 172) are all OUTSIDE finding 5's enumerated docLocation (':38-46, :163'). They sit in the decision table (§2), §4.2 data flow tail, §5 risk table, §9 DoD, and the changelog. The audit finding 5 scoped its fix narrowly to §3 and §10; the changelog (157, 172) is historical record that should not be rewritten. Per the "no scope creep" rule, I'll leave these ? they were not flagged by the audit, and the contracts they describe still hold (only the line numbers drifted, and these specific lines weren't in the worklist).

All 6 stale findings are now addressed. Let me do a final verification of the full diff to confirm the edits are clean and honest.

67. user (2026-06-22T09:50:27.070Z)

```
diff --git a/docs/PER_RALLY_SELECTION.md b/docs/PER_RALLY_SELECTION.md
index 25e6024..1afae68 100644
--- a/docs/PER_RALLY_SELECTION.md
+++ b/docs/PER_RALLY_SELECTION.md
@@ -3,13 +3,13 @@
> **Status:** `IN PROGRESS` ? **MVP / M0 (P1?P5) COMPLETE 2026-06-20** (pick ? query ? select ?
compile ? download, on existing engine endpoints). **Validated live e2e against a REAL engine
(Gemini) via the in-UI bucket?reel CUJ ? `DRY_RUN_CUJ.md` §9, 2026-06-21** (supersedes the
original stub-engine verification). Design **APPROVED by owner 2026-06-20**. Downstream
milestones M1?M5 (multi-video auth, per-rally preview endpoint, corpus loop, localized NL)
remain owner-gated. . **Owner:** Avi . **Created:** 2026-06-20 . **Last updated:** 2026-06-21
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
-> **Relation:** second major tracked work item in the `khelsutra` repo (peer of
`REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
+> **Relation:** second major tracked work item in the `khelsutra` repo (peer of
`docs/REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
> **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine code
this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
```

0. TL;DR

-Build the **pick-rallies ? reel** surface as a **hybrid**: a selectable rally **list/grid** is the workhorse; a **deterministic query bar** ("rallies over 10s, longest first") pre-selects into the **same** selection set; a sticky footer compiles the selection into one downloadable MP4. The entire MVP runs on **existing engine endpoints** ? `POST /api/query` ? `POST /api/compile` ? `GET /api/highlights` `[verified backend/main.py:331-340, 342-396, 307-328]` ? with **zero new engine intelligence and zero new endpoints**. The single most important caveat: queries and chips are scoped to **length / count / order / shot-count only**; shot-type, player, and score are **roadmap-only** (no detector, no column) and must not be faked. The biggest UX-vs-effort tension: **no per-rally preview is possible today** ? only the **final compiled reel** can stream ? so MVP selects on text metadata, and visual preview/timeline is a deliberate later milestone.

+Build the **pick-rallies ? reel** surface as a **hybrid**: a selectable rally **list/grid** is the workhorse; a **deterministic query bar** ("rallies over 10s, longest first") pre-selects into the **same** selection set; a sticky footer compiles the selection into one downloadable MP4. The entire MVP runs on **existing engine endpoints** ? `POST /api/query` ? `POST /api/compile` ? `GET /api/highlights` `[verified backend/main.py::{query\_play\_segments, compile\_highlights, download\_highlight}]` ? with **zero new engine intelligence and zero new endpoints**. The single most important caveat: queries and chips are scoped to **length / count / order / shot-count only**; shot-type, player, and score are **roadmap-only** (no detector, no column) and must not be faked. The biggest UX-vs-effort tension: **no per-rally preview is possible today** ? only the **final compiled reel** can stream ? so MVP selects on text metadata, and visual preview/timeline is a deliberate later milestone.

1. Problem & goal

Today the alpha can detect rallies and stitch a reel, but reel-building is either a **silent stitching.min_shot_count threshold** `[verified backend/main.py:362-383]` or a static checkbox list whose data plumbing is even broken ? the bundled static UI reads `data.rallies` while the API returns `play\_segments`, so it renders nothing `[verified backend/api/static/app.js:344 vs backend/main.py:340]`. The owner's framing is **dynamic per-video selection, not a fixed threshold**: a coach should say "give me the long rallies from this session" and get exactly those, then fine-tune by hand.

@@ -38,18 +38,18 @@ Today the alpha can detect rallies and stitch a reel, but reel-building is

either

3. Foundation ? what the engine gives us today `[verified this session]`
The whole design rests on the engine's real, already-shipped contract. Read against the engine repo (`badminton-highlight-indexer`):

```
-- **The rally store is `play_segments`** ? columns `id, video_id, start_time, end_time,
duration, server, receiver, metadata` `[verified database.py:124-135]`. `id` is the integer the
UI selects and passes to compile; `start_time/end_time/duration` are REAL seconds (duration
computed at insert).
-- **Listing rallies = `POST /api/query {video_id}` ? `{play_segments:[...]}`** ? the *only*
rally-list path today; it JSON-parses `metadata` and joins `events` `[verified main.py:331-340,
database.py:427-475]`.
-- **Building a reel = `POST /api/compile {video_id, segment_ids:[int], output_filename}`** ?
loads `videos.filepath`, filters all segments to the requested `segment_ids`, applies the
`min_shot_count` floor, extracts `time_pairs=[(start,end)]`, and calls
`VideoCompiler.compile_highlights(raw_path, segments, name)` `[verified main.py:342-396,
compiler.py:303]`, which **merges overlaps** then per-clip re-encodes ? concat ? one MP4
`[verified compiler.py:311-317]`. Selection order/duplicates are therefore safe.
-- **Downloading = `GET /api/highlights/{video_id}/{filename}`** ? streams the compiled MP4.
Compile returns a **server-local filepath, NOT a URL** `[verified main.py:394]`, so the SPA
constructs this download URL itself. The compiled reel is the **only streamable video that
exists today** `[verified main.py:307-328]`.
-- **The floor knob is readable** ? `GET /api/config` exposes `stitching.min_shot_count`
`[verified main.py:121-123]`, so the UI can warn before a selected rally is silently dropped.
-- **Job entry** ? `GET /api/jobs/{job_id}` returns `result.video_id` `[verified
main.py:274-293]`, which is how the MVP reaches a video without a (nonexistent) list-videos
endpoint.
+- **The rally store is `play_segments`** ? columns `id, video_id, start_time, end_time,
duration, server, receiver, metadata` `[verified database.py CREATE TABLE play_segments]`. `id`
is the integer the UI selects and passes to compile; `start_time/end_time/duration` are REAL
seconds (duration computed at insert).
+- **Listing rallies = `POST /api/query {video_id}` ? `{play_segments:[...]}`** ? the *only*
rally-list path today; it JSON-parses `metadata` and joins `events` `[verified
main.py::query_play_segments, database.py::query_play_segments]`.
+- **Building a reel = `POST /api/compile {video_id, segment_ids:[int], output_filename}`** ?
loads `videos.filepath`, filters all segments to the requested `segment_ids`, applies the
`min_shot_count` floor, extracts `time_pairs=[(start,end)]`, and calls
`VideoCompiler.compile_highlights(raw_path, segments, name)` `[verified
main.py::compile_highlights, compiler.py::compile_highlights]`, which **merges overlaps** then
per-clip re-encodes ? concat ? one MP4 `[verified compiler.py::merge_overlapping]`. Selection
order/duplicates are therefore safe.
+- **Downloading = `GET /api/highlights/{video_id}/{filename}`** ? streams the compiled MP4.
Compile now returns **both** a server-local `filepath` **and** a ready-to-use `download_url`
(`/api/highlights/...`) `[verified main.py::compile_highlights]`, so the SPA may use
`download_url` directly; the hand-built-URL fallback is now optional, not required. The compiled
reel is the **only streamable video that exists today** `[verified
main.py::download_highlight]`.
+- **The floor knob is readable** ? `GET /api/config` exposes `stitching.min_shot_count`
`[verified main.py::get_config]`, so the UI can warn before a selected rally is silently
dropped.
+- **Job entry** ? `GET /api/jobs/{job_id}` returns `result.video_id` `[verified
main.py::get_job]`, which is how the MVP reaches a video; a list-videos endpoint (`GET
/api/videos`) has since shipped ($4.3).
```

```
**Honest fields vs fabricated/roadmap (critical for the UI):**
- **Honest today** (default `default_segmenter: "gemini"`, per `config.json` `[verified]`):
`start_time`, `end_time`, `duration`, and `metadata.shot_count` (Gemini's
`shuttle_exchange_count`, fallback `max(3, int(dur/0.8))`) + `metadata.shot_count_confidence` (a
**string**, often `"Unknown"`) `[verified gemini.py:470-485]`. On the gemini path
`server`/`receiver` are written `None` ("no fabricated data") `[verified gemini.py:480-487]`.
- **Fabricated ? do NOT surface:** the `motion` demo segmenter fills `server`/`receiver` via
`random.sample(...)` , a random `shot_count`, and random `events` (serve/smash/foul/winner)
`[verified motion.py:191-198]`. `query` returns these columns + `events`, but they are not
ground truth and must stay hidden.
- **Roadmap ? no column, no detector:** shot-type per shot, player identity, match/point score,
```

game/set grouping. None exist anywhere in the schema; do not consume them `[verified database.py whole-schema 122-148]`.

-- **Correction-loop columns exist but are dead:**

`candidate\_segments.human\_label/notes/confirmed\_segment\_id` are defined but no endpoint writes them `[verified database.py:166-168]` ? that's the M4 corpus extension (engine PR-4/B6, `UI\_ALPHA\_EXECUTION\_PLAN.md`).

+-- **Correction-loop columns exist but are dead:**

`candidate\_segments.human\_label/notes/confirmed\_segment\_id` are defined but no endpoint writes them `[verified database.py CREATE TABLE candidate\_segments]` ? that's the M4 corpus extension (engine PR-4/B6, `UI\_ALPHA\_EXECUTION\_PLAN.md`).

4. Design / architecture

@@ -70,7 +70,7 @@ Gemini segmenter (process time) writes `play_segments {id, start/end_time, durat

? all input modes mutate the **same** `selectedIds`: (a) `parseQuery()` predicate over duration/ordinal/shot_count; (b) checkbox toggle + bulk ops; (c) *(post-MVP)* timeline band ? footer recomputes count + ?duration **client-side**, and warns for any selected rally with `shot\_count < stitching.min\_shot\_count` `[main.py:362-383]`

? **Build** ? `POST /api/compile {video\_id, segment\_ids: [...selectedIds], output\_filename}` ? stitcher de-dupes via `\_merge\_overlapping` `[compiler.py:311-317]`, cuts `time\_pairs` from `videos.filepath`, re-encodes per clip ? concat ? one MP4 `[main.py:386-393, compiler.py:303]`
-? compile returns a **server-local** filepath, not a URL `[main.py:394]` ? SPA builds `GET /api/highlights/{video\_id}/{output\_filename}` to preview in `` + download `[main.py:307-328]`.

+? compile returns **both** a server-local `filepath` **and** a ready-to-use `download\_url` (`/api/highlights/...`) `[main.py::compile\_highlights]` ? the SPA may use `download\_url` directly (or build `GET /api/highlights/{video\_id}/{output\_filename}` itself) to preview in `` + download `[main.py::download\_highlight]`.

****4.3 Engine API additions (none for MVP; tagged against real gaps)****

| Endpoint | Purpose | Status |

@@ -80,10 +80,10 @@ Gemini segmenter (process time) writes `play_segments {id, start/end_time, durat

`GET /api/highlights/{video\_id}/{filename}`	Stream/download compiled reel	**exists**
`GET /api/config`	Read `stitching.min\_shot\_count` for the warning	**exists**
`GET /api/jobs/{job\_id}`	Carry `result.video\_id` into the picker	**exists**
-	`POST /api/compile` ? add `download\_url`	Return `/api/highlights/...` so SPA stops hand-building it (additive, low-risk; today returns only a filepath `main.py:394`)
-	`GET /api/health`	Engine-offline banner ping the SPA is specified to use (`HOSTING\_PLAN.md:42`)
+	`POST /api/compile` ? add `download\_url`	Returns `/api/highlights/...` so the SPA needn't hand-build it (compile now returns both `filepath` **and** `download\_url` `[verified main.py::compile\_highlights]`)
+	`GET /api/health`	Engine-offline banner ping the SPA is specified to use (`HOSTING\_PLAN.md:42`) ? shipped `[verified main.py::health]`
`GET /api/videos/{video\_id}/rallies`	RESTful/cacheable alias returning the same `{play\_segments}` shape	**new**
-| `GET /api/videos` | List a user's videos for a Match picker (only `get\_video(id)` exists, `database.py:418`) ? needs DB enumerate + identity scoping | **new** |
+| `GET /api/videos` | List videos for a Match picker ? the enumerate endpoint shipped `[verified main.py::list\_videos, database.py::list\_videos]`; only identity/owner-scoping remains gated | **exists** |

| `GET /api/videos/{video\_id}/source` (Range) *or* `/clip?start=&end=` | Stream source/per-span video so a rally previews before selection ? neither exists | **new** |

| `GET /api/videos/{video\_id}/rallies/{segment\_id}/thumbnail` | Per-rally poster still (no thumbnail column; WASB frames are transient) | **new** |

| `PATCH /api/segments/{id}` | Persist accept/reject/nudge ?

`candidate\_segments.human\_label/notes/...` (columns exist, unwritten ? corpus loop) | **new** |

@@ -113,7 +113,7 @@ No silent drops: unsupported concepts (`query.unavailable\_request` + `dropped_un

| No match-duration field ? a timeline must derive `total = max(end\_time)`, mis-rendering pre/post dead air | Med | Timeline deferred to M3; never over-promise it as a true scrubber until source streams |

| `shot\_count` is an **estimate** (`shuttle\_exchange\_count` w/ `max(3,int(dur/0.8))` fallback)


```
`[gemini.py:470-485]` | Med | Label "approx."; confidence shown as the raw string, not a meter |  
| No auth / user scoping anywhere (CORS `*`, `videos.user_id` never set/filtered) | High for  
multi-tester | Fine for single-box local alpha; a ship-blocker for any shared host ? gate `GET  
/api/videos` behind PR-3 scoping |  
-| compile returns a filepath not a URL `[main.py:394]` | Low | Client builds the URL for MVP;  
`download_url` field (M1) removes the brittleness |  
+| compile returns a server-local filepath the browser can't reach  
`[main.py::compile_highlights]` | Low | Resolved: compile now also returns a ready-to-use  
`download_url`; the client may use it directly (the hand-built-URL fallback still works) |  
| Surface sprawl (query + grid + future timeline + future correction) | Med | Deliberate IA:  
query+grid = one screen; correction = separate ? corpus screen (owner call, $8) |
```

```
## 6. Honest limits ? what this does NOT do  
@@ -127,12 +127,12 @@ Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One smal  
| P0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | No | ? |  
[#11](https://github.com/avidullu/khelsutra/pull/11) |  
| P1 | SPA scaffold (Vite+React+TS+Tailwind) in `web/` + typed API client; dev proxy to engine  
`:8000` | P0 | No | ? | [#14](https://github.com/avidullu/khelsutra/pull/14) |  
| P2 | RallyList/Grid (honest fields only) + `useRallySelection()` hook (unit-tested) | P1 | No  
| ? | [#21](https://github.com/avidullu/khelsutra/pull/21) |  
-| P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable tokens)  
| P2 | No | ? | (this PR) |  
-| P4 | SelectionFooter + `min_shot_count` warning (from `GET /api/config`) | P2 | No | ? |  
(this PR) |  
-| P5 | Build ? `POST /api/compile` ? client-built download URL + `` preview of the reel  
| P2 | No | ? | (this PR) |  
-| P6 | *(engine repo)* `GET /api/health` + offline banner; `/api/compile` returns  
`download_url` | ? | No | ? | ? |  
+| P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable tokens)  
| P2 | No | ? | [#28](https://github.com/avidullu/khelsutra/pull/28) |  
+| P4 | SelectionFooter + `min_shot_count` warning (from `GET /api/config`) | P2 | No | ? |  
[#30](https://github.com/avidullu/khelsutra/pull/30) |  
+| P5 | Build ? `POST /api/compile` ? client-built download URL + `` preview of the reel  
| P2 | No | ? | [#31](https://github.com/avidullu/khelsutra/pull/31) |  
+| P6 | *(engine repo)* `GET /api/health` + offline banner; `/api/compile` returns  
`download_url` | ? | No | ? | engine halves shipped (`main.py::health`, compile `download_url`);  
SPA offline banner + `download_url` consumption still pending |  
| P7 | *(engine repo)* `GET /api/videos/{id}/rallies` RESTful alias | ? | No | ? | ? |  
-| P8 | *(engine repo)* `GET /api/videos` + PR-3 identity scoping ? A1 Matches screen | P7 | ?  
owner | ? | ? |  
+| P8 | *(engine repo)* `GET /api/videos` (listing endpoint **shipped** ?  
`main.py::list_videos`, `database.py::list_videos`) + PR-3 identity scoping (still gated) ? A1  
Matches screen | P7 | ? owner | ? | ? |  
| P9 | *(engine repo)* source/clip stream + timeline + per-rally preview/thumbnail | P8 | ?  
owner | ? | ? |  
| P10 | *(engine repo)* `PATCH /api/segments/{id}` corpus loop (PR-4/B6) ? A4 Rally Review | P9  
| ? owner | ? | ? |
```

```
@@ -160,8 +160,8 @@ Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One smal  
- [x] $7 tracker + this DoD updated. *(Doc stays live for the owner-gated M1?M5 engine  
milestones; not archived ? only the MVP/M0 slice is complete.)*
```

```
## 10. References  
-**Engine code anchors `[verified this session]`:** `backend/main.py:121-123` (config),  
`:274-293` (jobs), `:307-328` (highlights download), `:331-340` (query), `:342-396` (compile +  
`min_shot_count` floor + filepath-not-URL); `backend/infrastructure/database.py:124-135`  
(play_segments), `:166-168` (dead correction columns), `:418` (no list-videos), `:427-475`  
(query read path + events join); `backend/pipeline/segmenters/gemini.py:461-498, 470-485`  
(honest shot_count, server/receiver=None); `backend/pipeline/segmenters/motion.py:191-198`  
(fabricated fields); `backend/pipeline/stitcher/compiler.py:80-105, 303, 311-317` (merge/de-dupe  
+ entypoint); `backend/api/static/app.js:344` (`data.rallies` bug); `config.json`  
(`default_segmenter: "gemini"`).  
-**Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`;  
`HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now,  
model punted). **This repo:** `web/README.md`, `REQUEST_ACCESS_FORM.md`.
```

```

***Engine code anchors `[verified this session]` (symbol-based ? engine line numbers drift
between PRs):** `backend/main.py::get_config` (config), `::get_job` (jobs),
`::download_highlight` (highlights download), `::query_play_segments` (query),
`::compile_highlights` (compile + `min_shot_count` floor + now returns `download_url`),
`::health` (offline ping), `::list_videos` (videos list ? shipped);
`backend/infrastructure/database.py CREATE TABLE play_segments` (play_segments), `CREATE TABLE
candidate_segments` (dead correction columns `human_label/notes/confirmed_segment_id`),
`::list_videos` (videos enumerate ? shipped), `::query_play_segments` (query read path + events
join); `backend/pipeline/segmenters/gemini.py` (honest shot_count, server/receiver=None);
`backend/pipeline/segmenters/motion.py` (fabricated fields);
`backend/pipeline/stitcher/compiler.py::{_merge_overlapping, compile_highlights}` (merge/de-dupe
+ entrypt); `backend/api/static/app.js:344` (`data.rallies` bug); `config.json`
(`default_segmenter: "gemini"`).
***Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`;
`HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now,
model punted). **This repo:** `web/README.md`, `docs/REQUEST_ACCESS_FORM.md`.

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```

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### Changelog

```

68. assistant (2026-06-22T09:50:34.639Z)

The diff is clean and surgical. All 6 stale findings are addressed. The milestone rollup at line 139 still says "M1 = P6?P7 (RESTful + reliability polish)" ? that's consistent with P6 being ? (partial).

One thing to note: P6 is now ? but the milestone rollup says "M0/MVP = P1?P5 ? COMPLETE" which is unchanged and correct. I deliberately left the §2 decision-table, §4.2 floor anchors, §5 risk rows 111-112, §9 DoD, and changelog line-number anchors untouched as they're outside the audited findings' scope (no scope creep) and the changelog is historical record.

Let me return the structured report.

69. user (2026-06-22T09:51:04.516Z)

Structured output provided successfully

70. assistant (2026-06-22T09:51:11.718Z)

I have completed the surgical staleness fixes for `docs/PER\_RALLY\_SELECTION.md`. All 6 audited stale findings were re-verified against the live engine repo (`badminton-highlight-indexer` origin/master, freshly fetched) and the live khelsutra repo, confirmed, and fixed with minimal, targeted edits (21 insertions / 21 deletions, no wholesale rewrite). Key outcomes:

- **download_url shipped** (findings 1/2): compile now returns both `filepath` and `download\_url` ? caveat reworded everywhere in scope; §4.3 status flipped to **exists**.
- **GET /api/health + GET /api/videos shipped** (findings 1/3): §4.3 statuses flipped to **exists**; corrected the false "only get_video(id) exists" claim.
- **P3/P4/P5 PRs** (finding 4): `(this PR)` placeholders backfilled with #28/#30/#31.
- **Engine anchor drift** (finding 5): converted brittle line numbers to symbol references in §3 and §10.
- **Doc path** (finding 6): `REQUEST\_ACCESS\_FORM.md` ? `docs/REQUEST\_ACCESS\_FORM.md` (both spots).

Honesty gates respected: P6 and P8 were set to ? (not ?) because their SPA-side / owner-scoping halves remain pending; the historical Changelog and out-of-scope line-number anchors were intentionally left untouched to avoid scope creep. No code changes were needed.